

Nature Coast Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

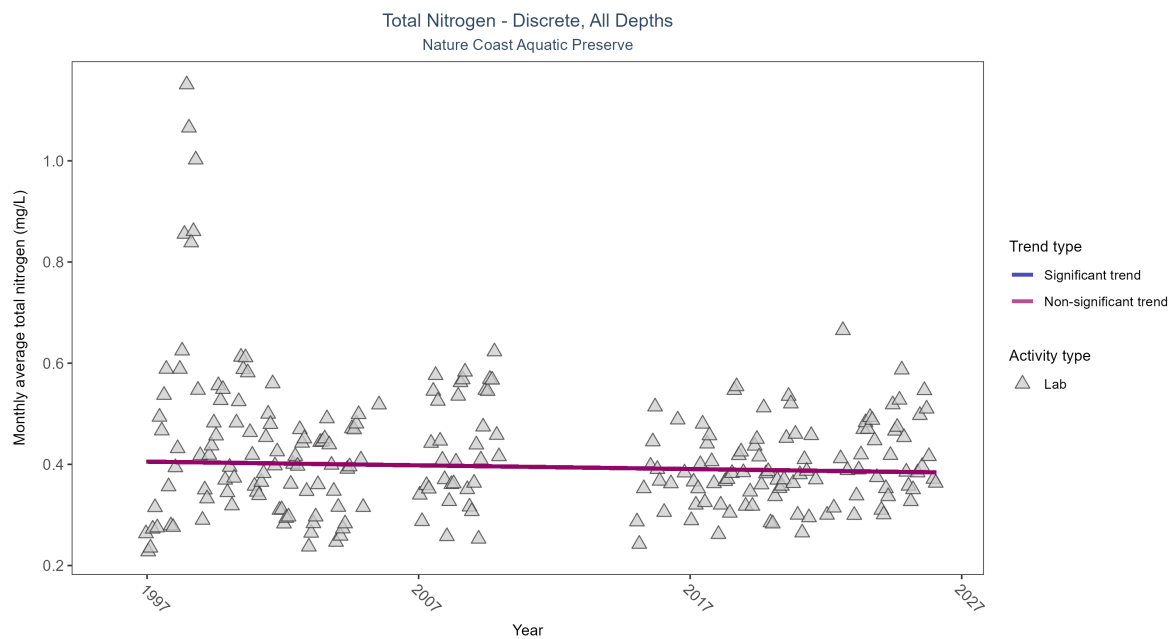


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Total nitrogen showed no detectable trend between 1996 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	7039	25	1996 - 2026	0.39	-0.07163	0.40605	-0.00073	0.0949

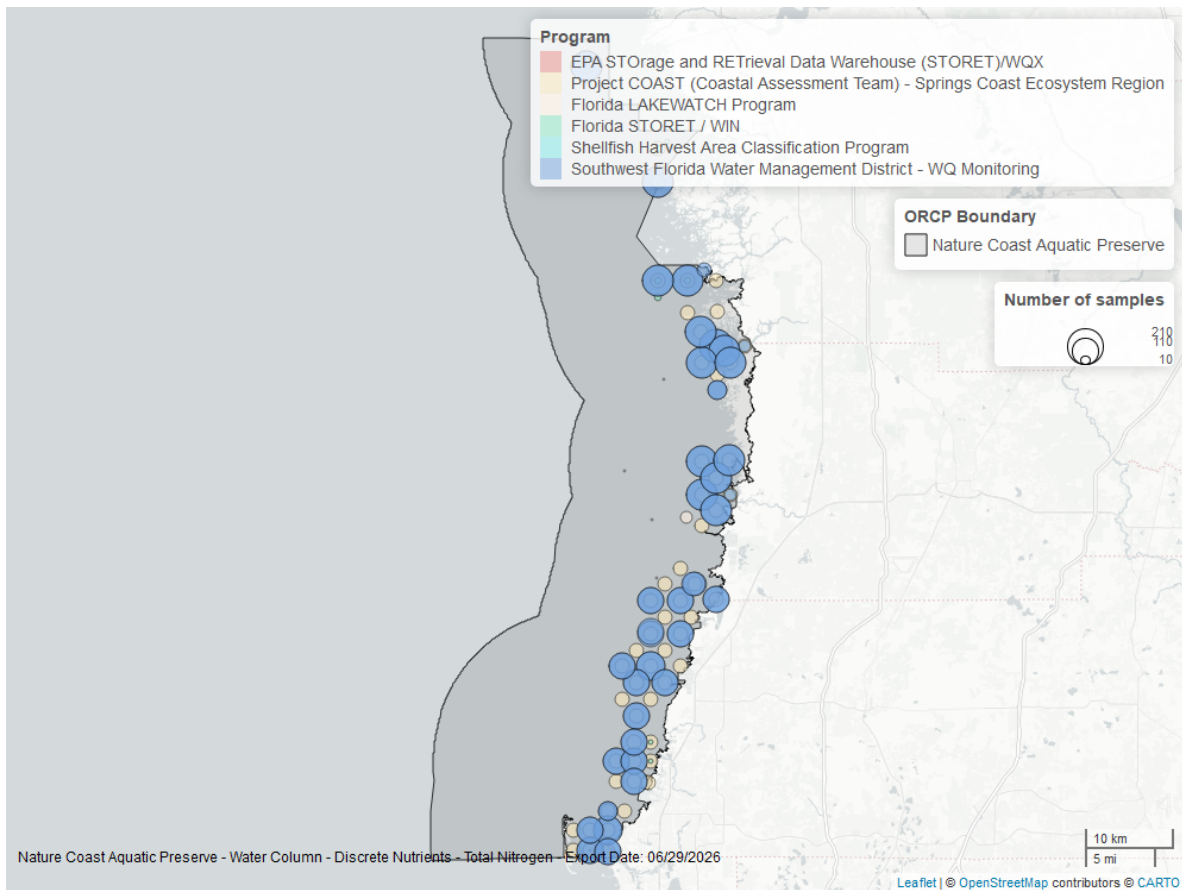


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

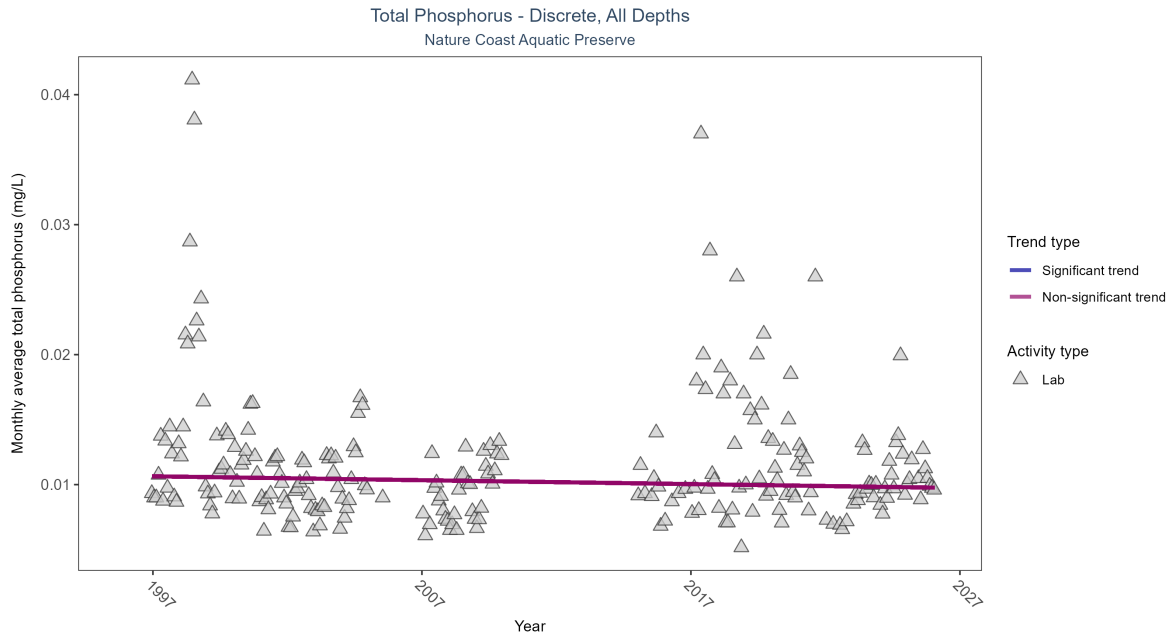


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Total phosphorus showed no detectable trend between 1996 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	7094	25	1996 - 2026	0.00855	-0.07942	0.01067	-0.00003	0.0701

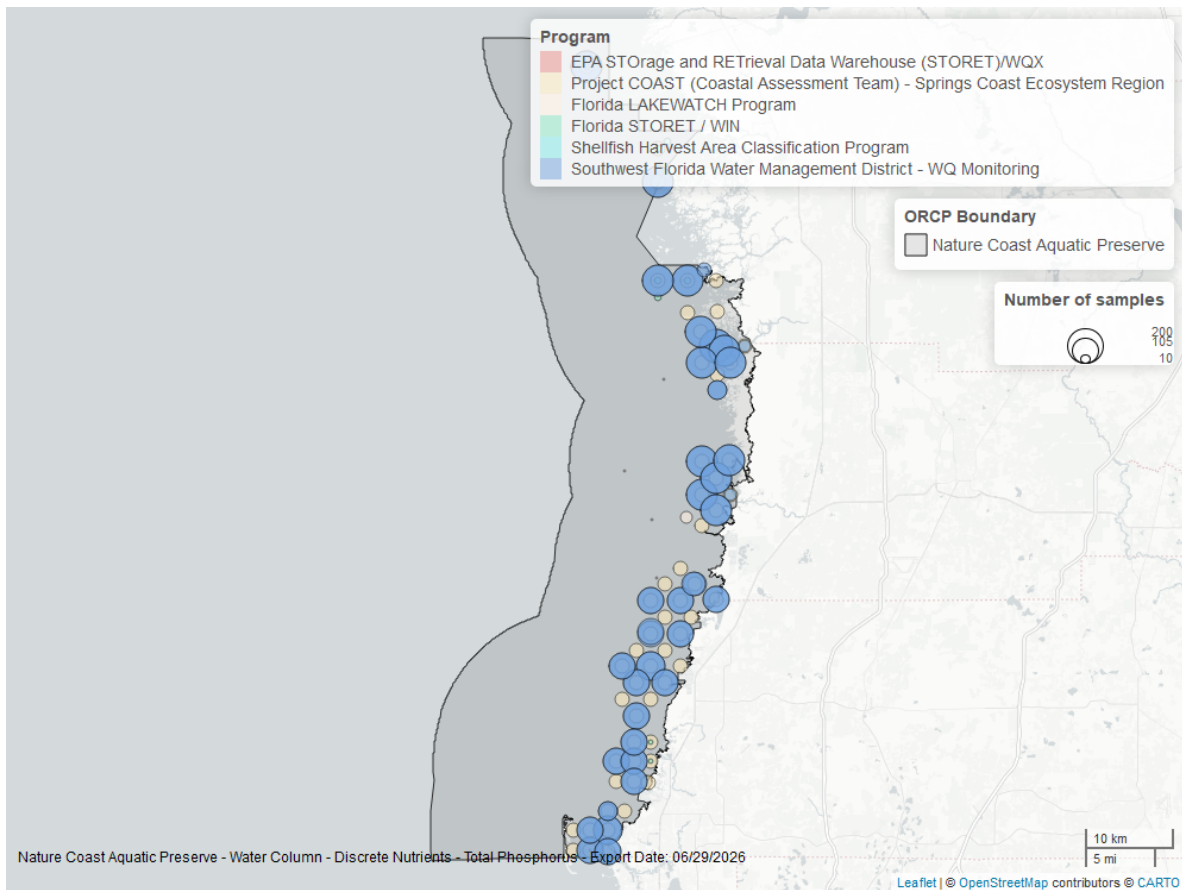


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

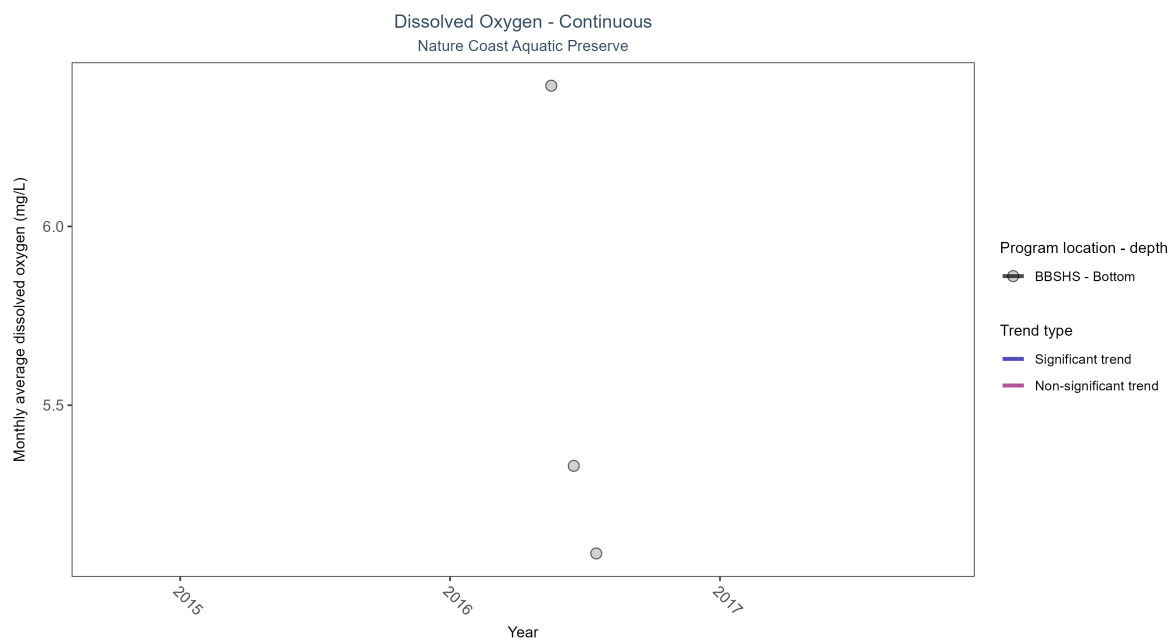


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSHS	Insufficient data	6555	1	2016 - 2016	5.7	-	-	-	-

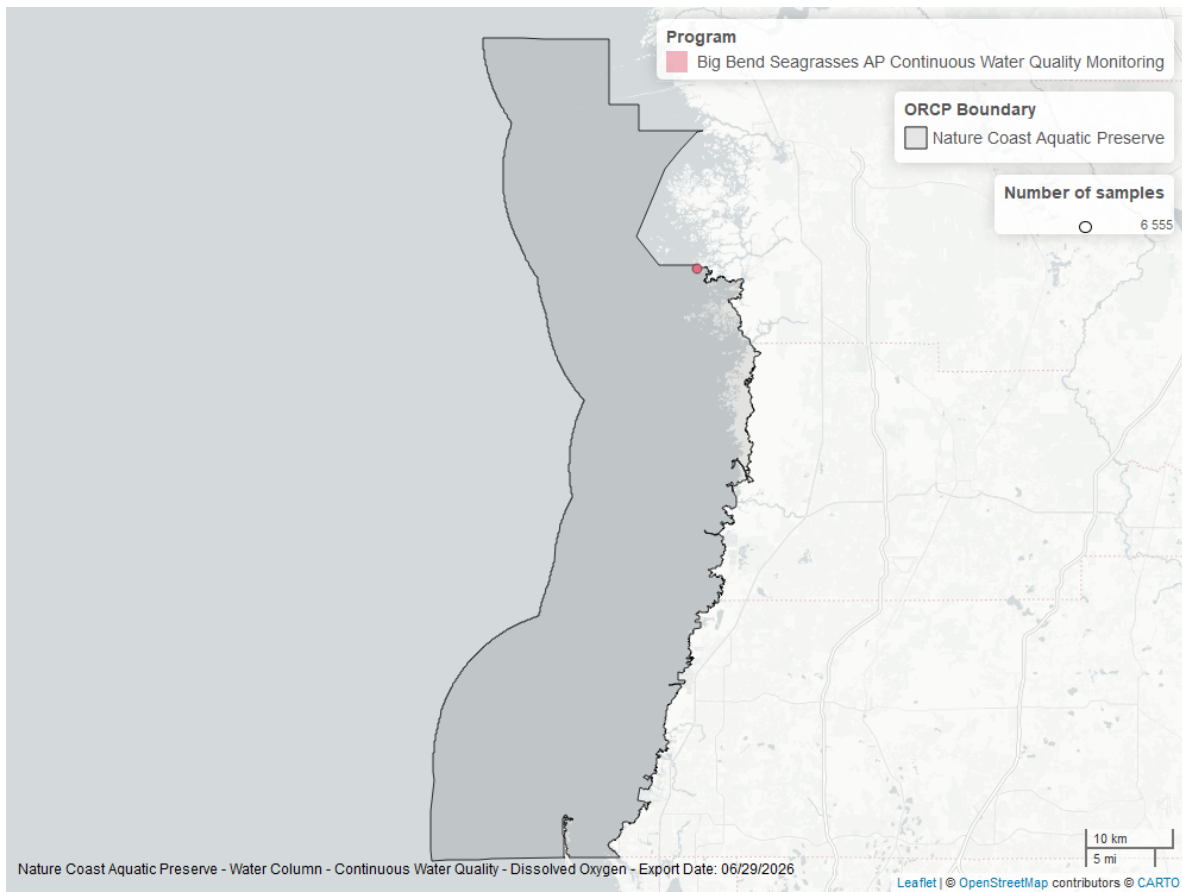


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

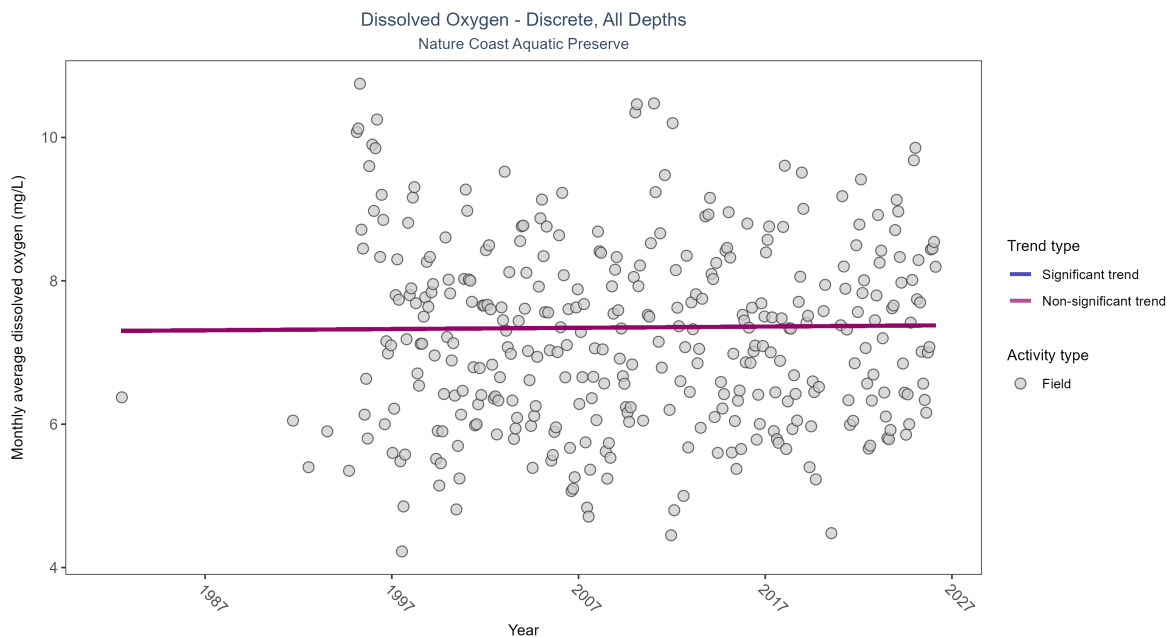


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Dissolved oxygen showed no detectable trend between 1982 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	10309	37	1982 - 2026	7.21	0.01471	7.30108	0.00175	0.6866

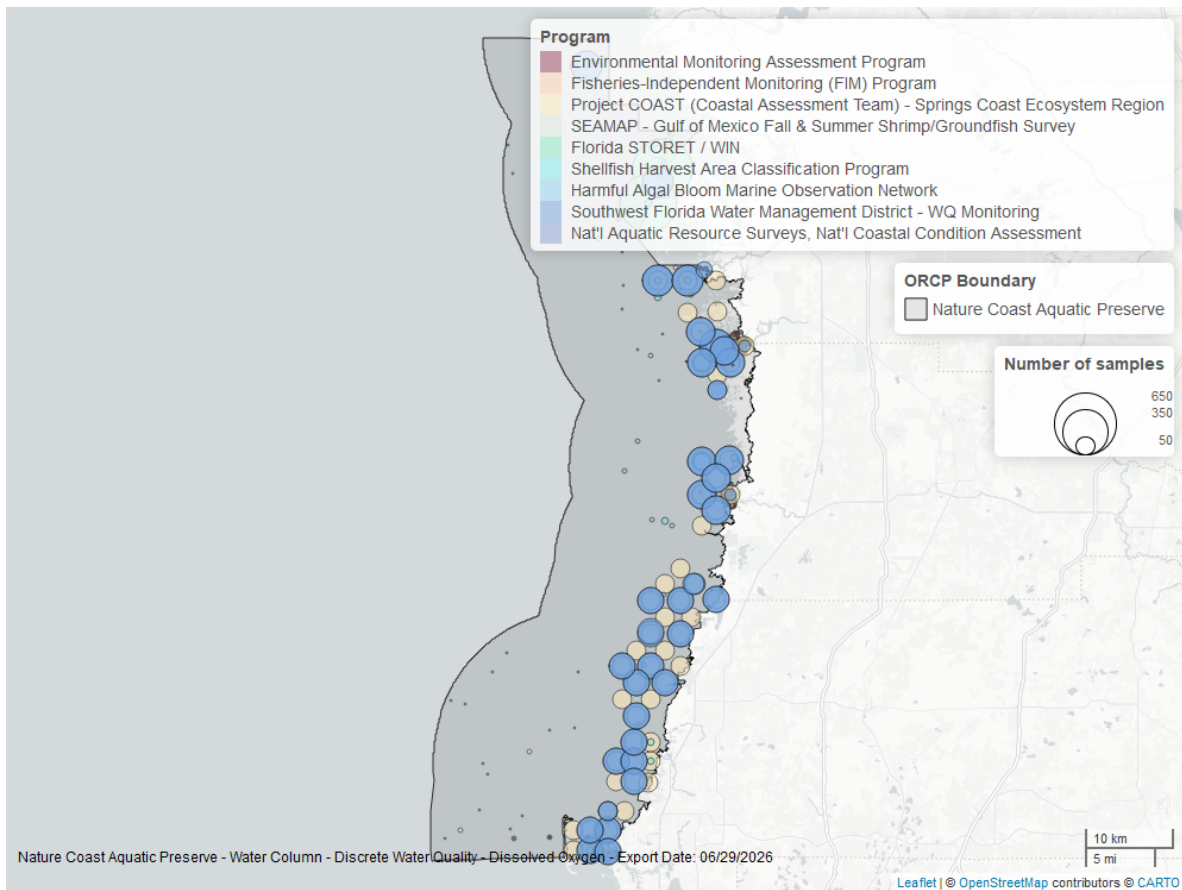


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

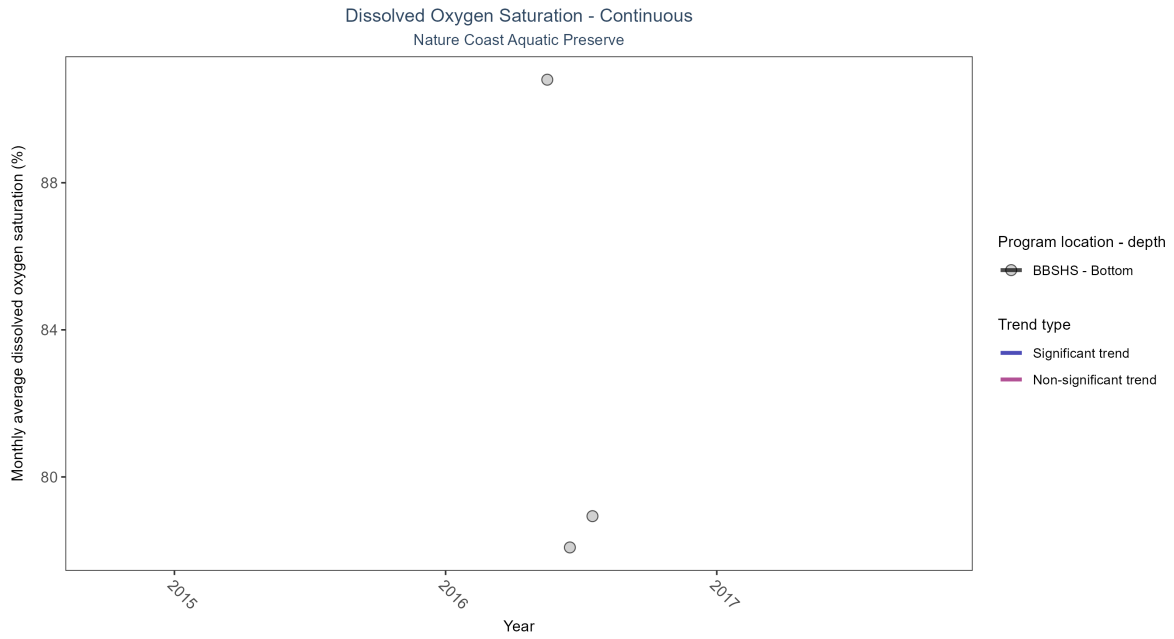


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSHS	Insufficient data	6555	1	2016 - 2016	82.6	-	-	-	-

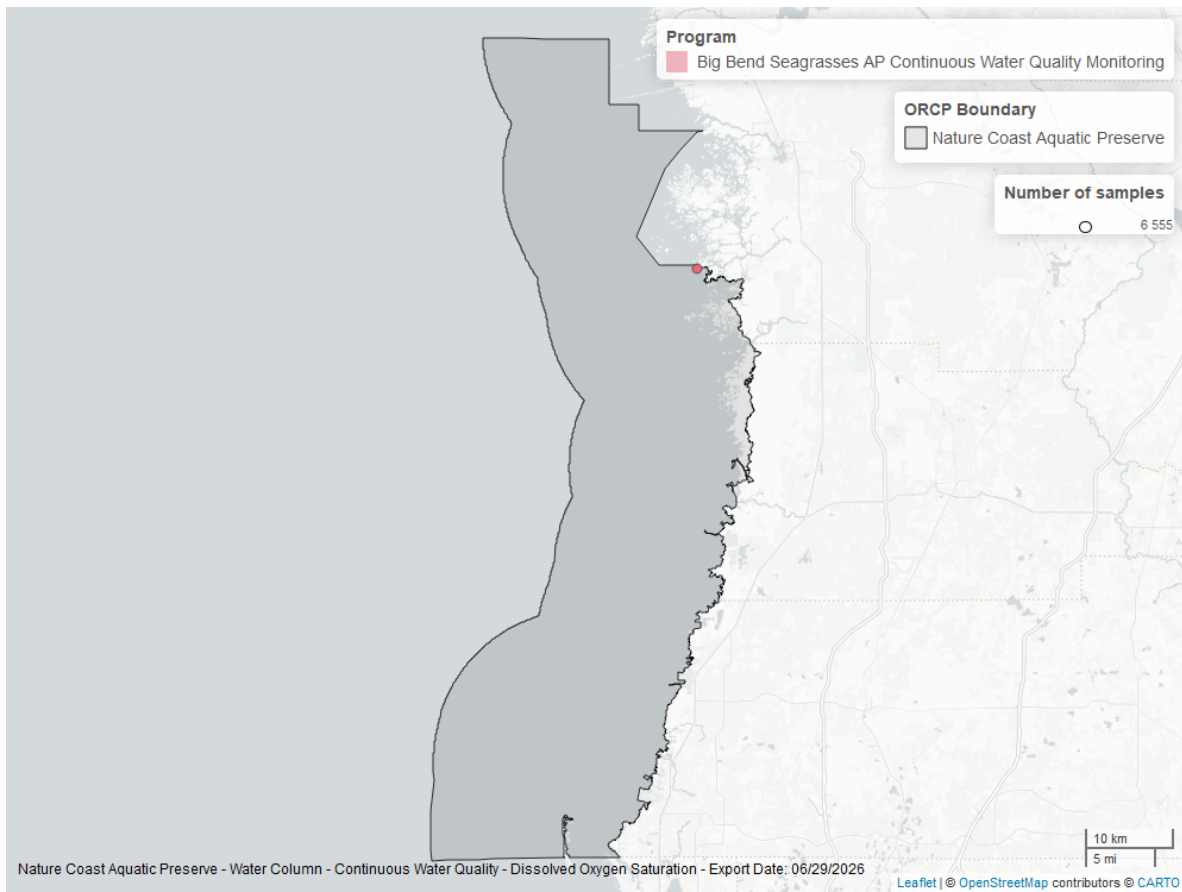


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

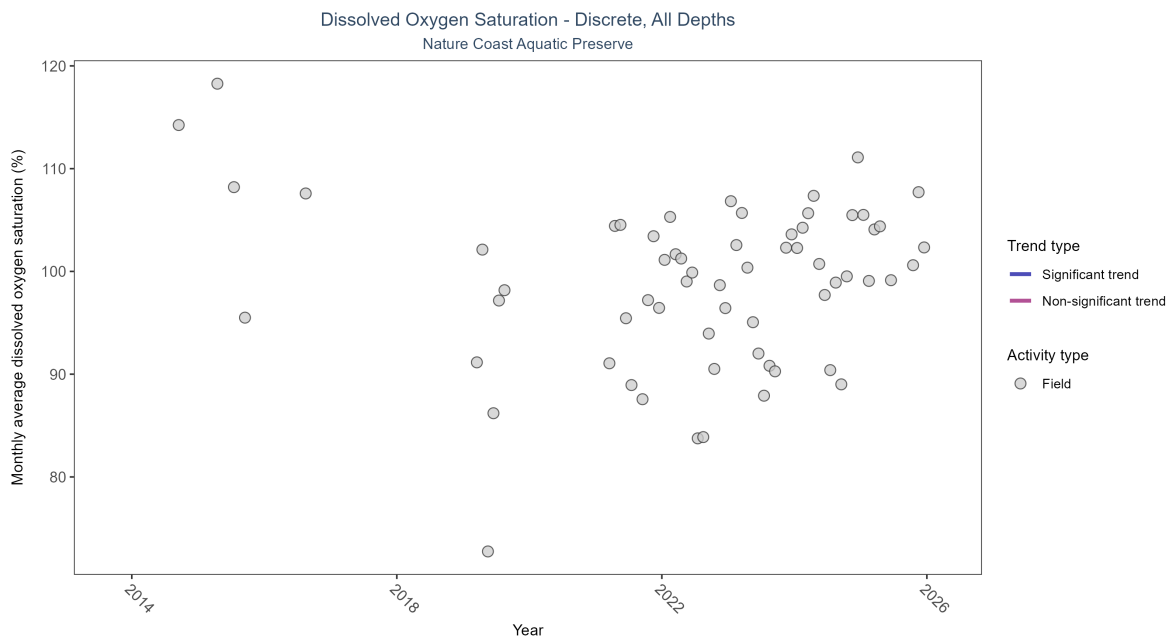


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: There was insufficient data to fit a model for dissolved oxygen saturation.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Insufficient data	3260	9	2014 - 2025	100.4	-	-	-	-

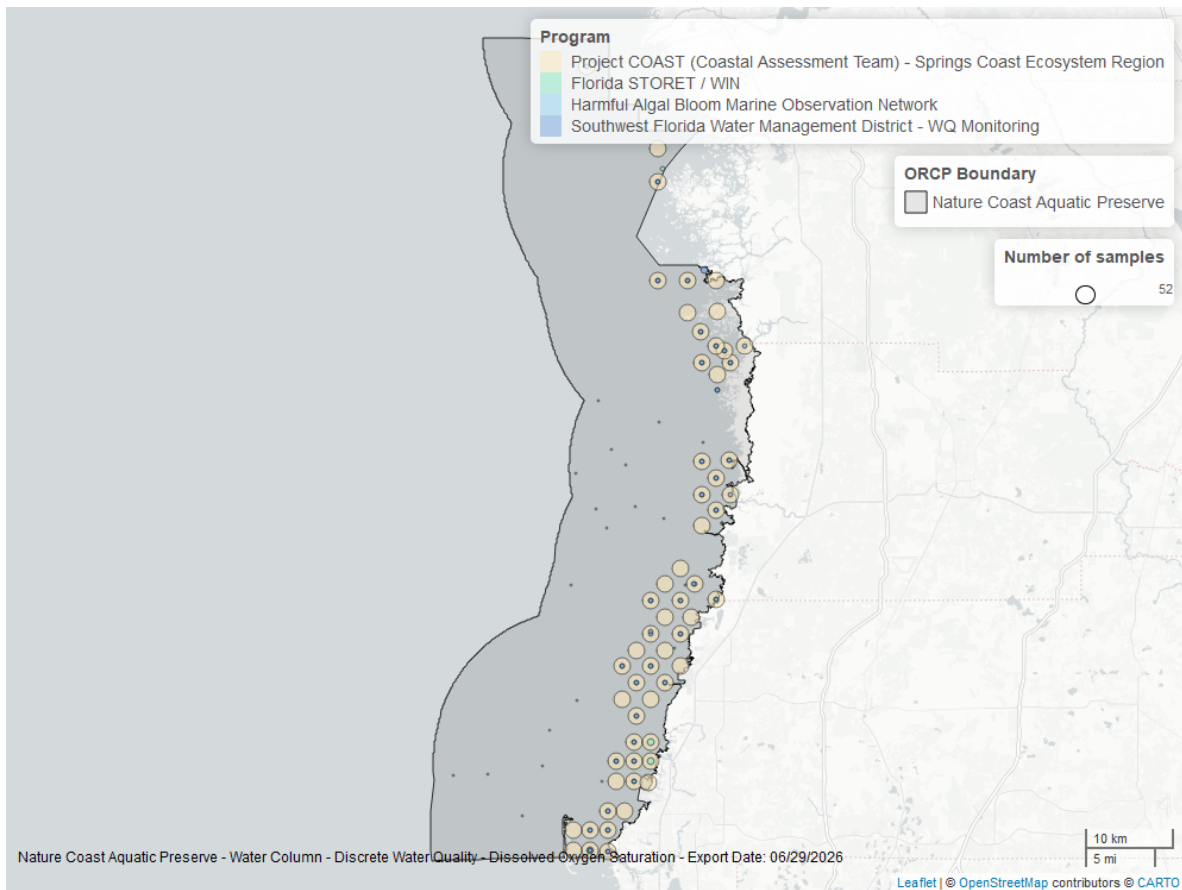


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

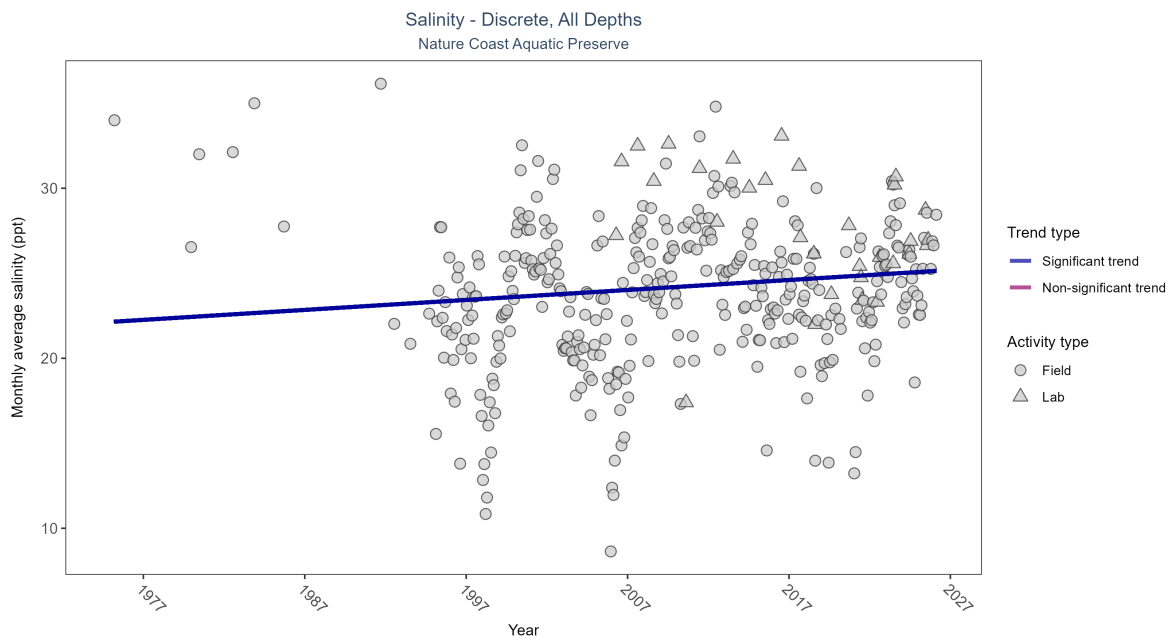


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Monthly average salinity increased by 0.06 ppt per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Increasing trend	11210	42	1975 - 2026	24.62	0.09253	22.14096	0.05862	0.0123

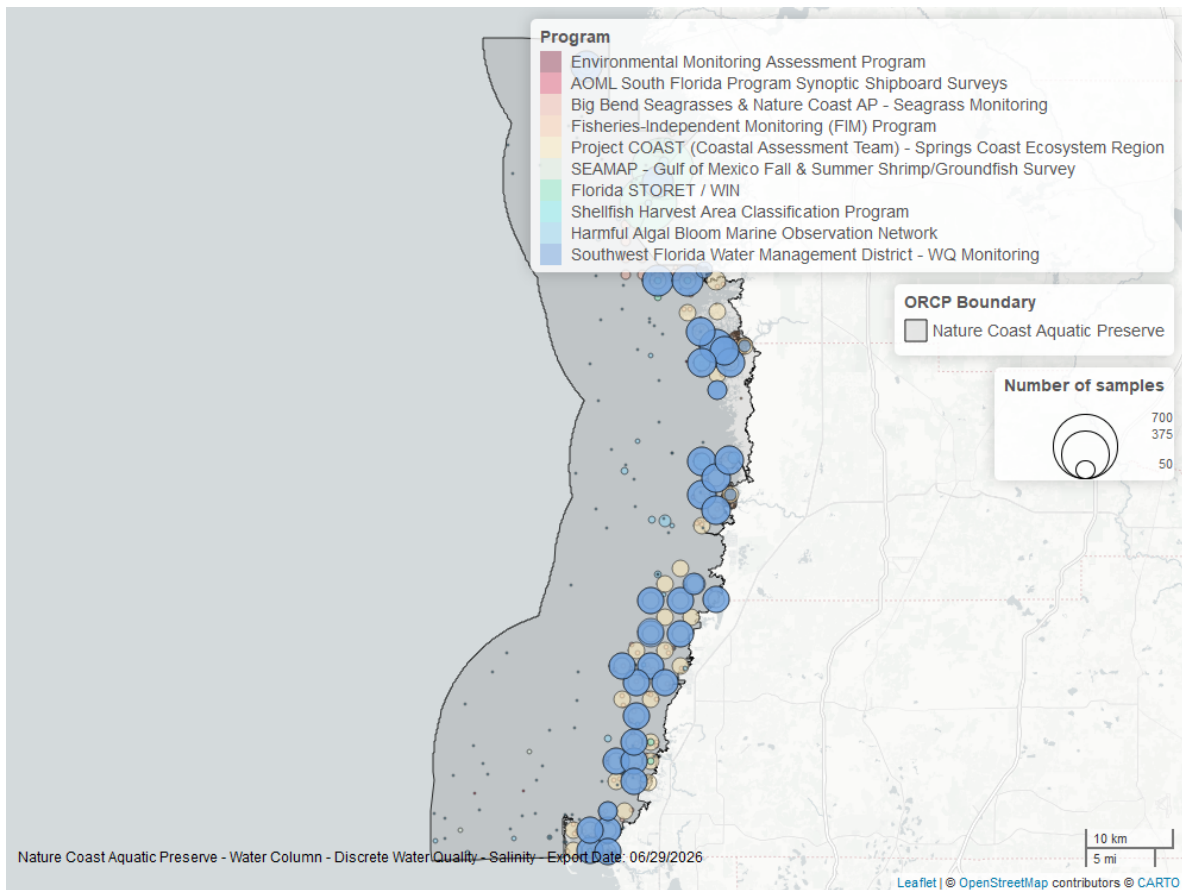


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

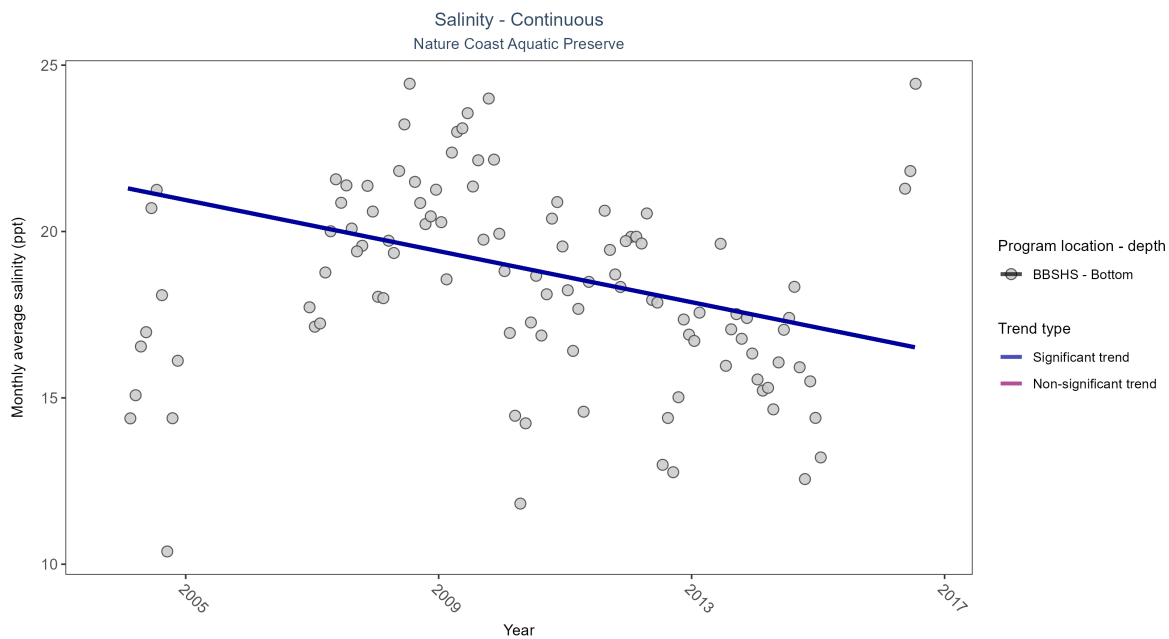


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: At one program location, monthly average salinity decreased by 0.38 ppt per year.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSHS	Decreasing trend	235670	12	2004 - 2016	18.4	-0.22	21.33	-0.38	0

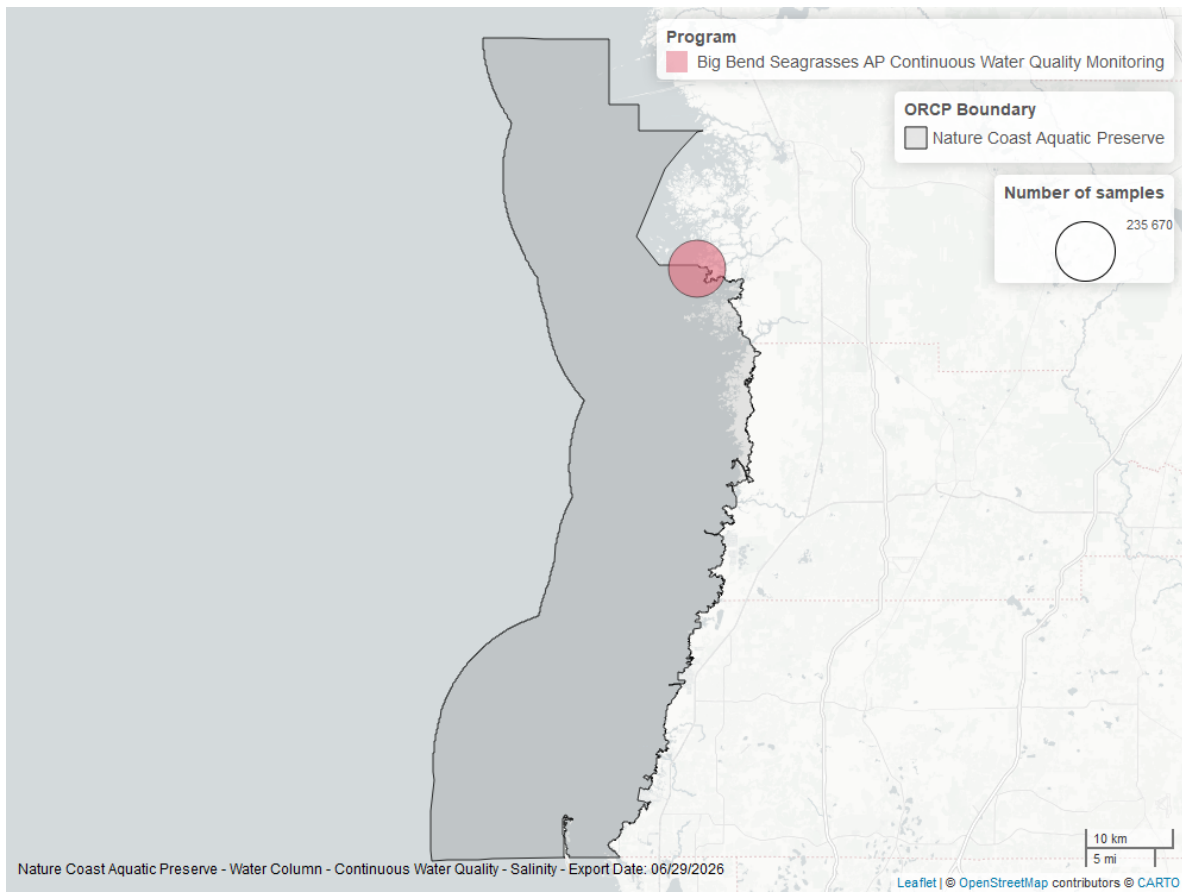


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

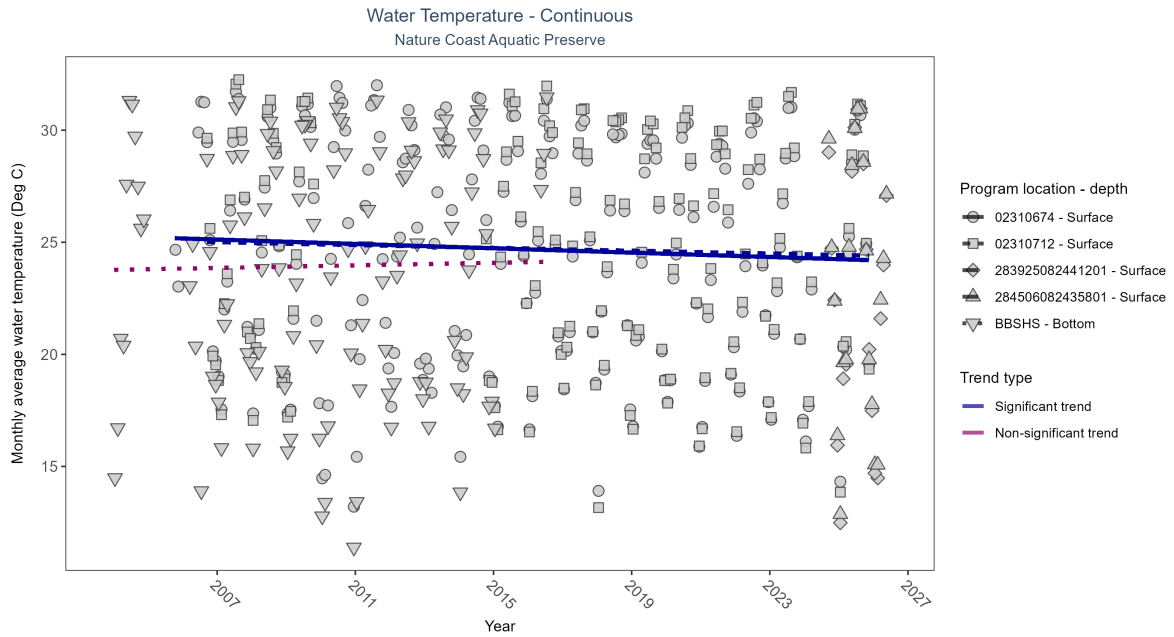


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: At two program locations, monthly average water temperature decreased by 0.03Deg C per year at one site and by 0.05Deg C per year at the other. No detectable change in monthly average water temperature was observed at one location. There was insufficient data to fit a model for two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02310674	Decreasing trend	9951	21	2005 - 2025	25.5	-0.17	25.22	-0.05	0
02310712	Decreasing trend	7075	16	2006 - 2025	25.4	-0.12	25.03	-0.03	0.04
283925082441201	Insufficient data	931	3	2024 - 2026	22.5	-	-	-	-
284506082435801	Insufficient data	1033	3	2024 - 2026	22.7	-	-	-	-
BBSHS	No detectable trend	244110	12	2004 - 2016	24.5	0.05	23.77	0.03	0.52

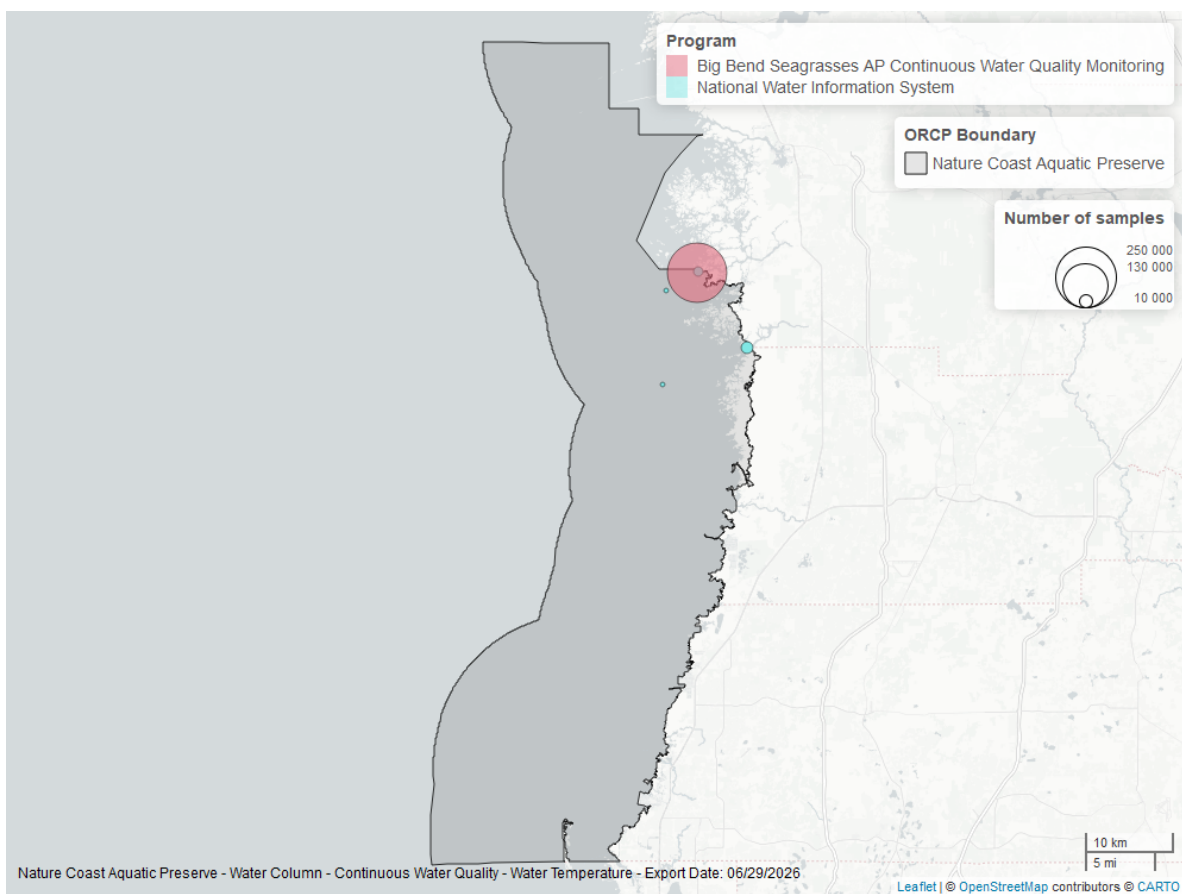


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

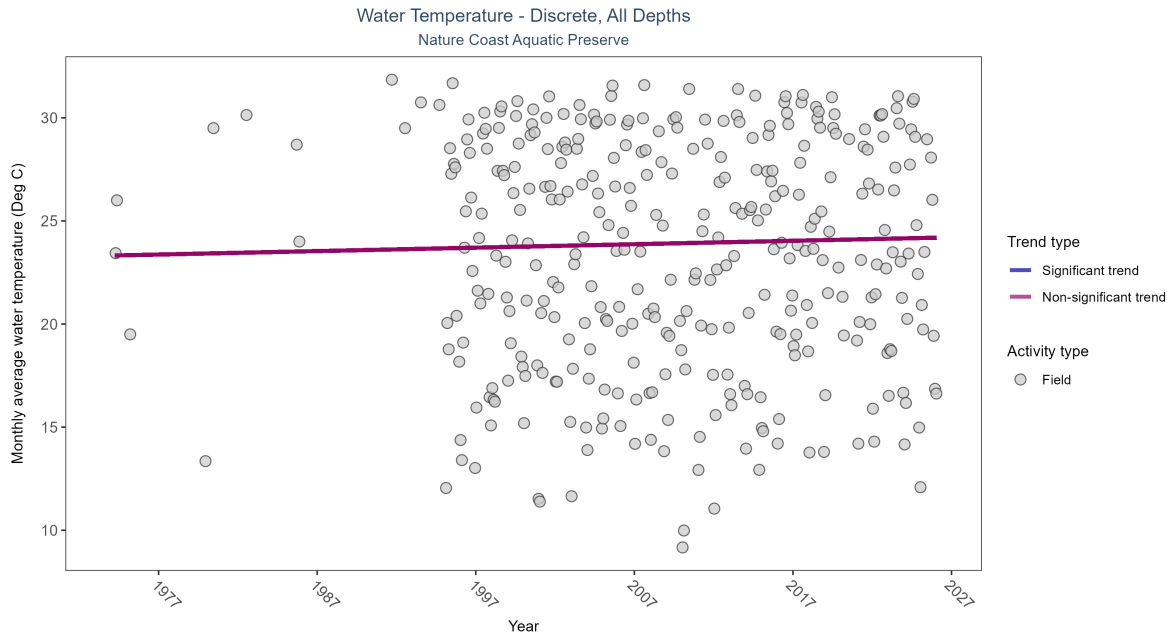


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 10: Water temperature showed no detectable trend between 1974 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	9150	42	1974 - 2026	24.1	0.07196	23.32112	0.01667	0.0611

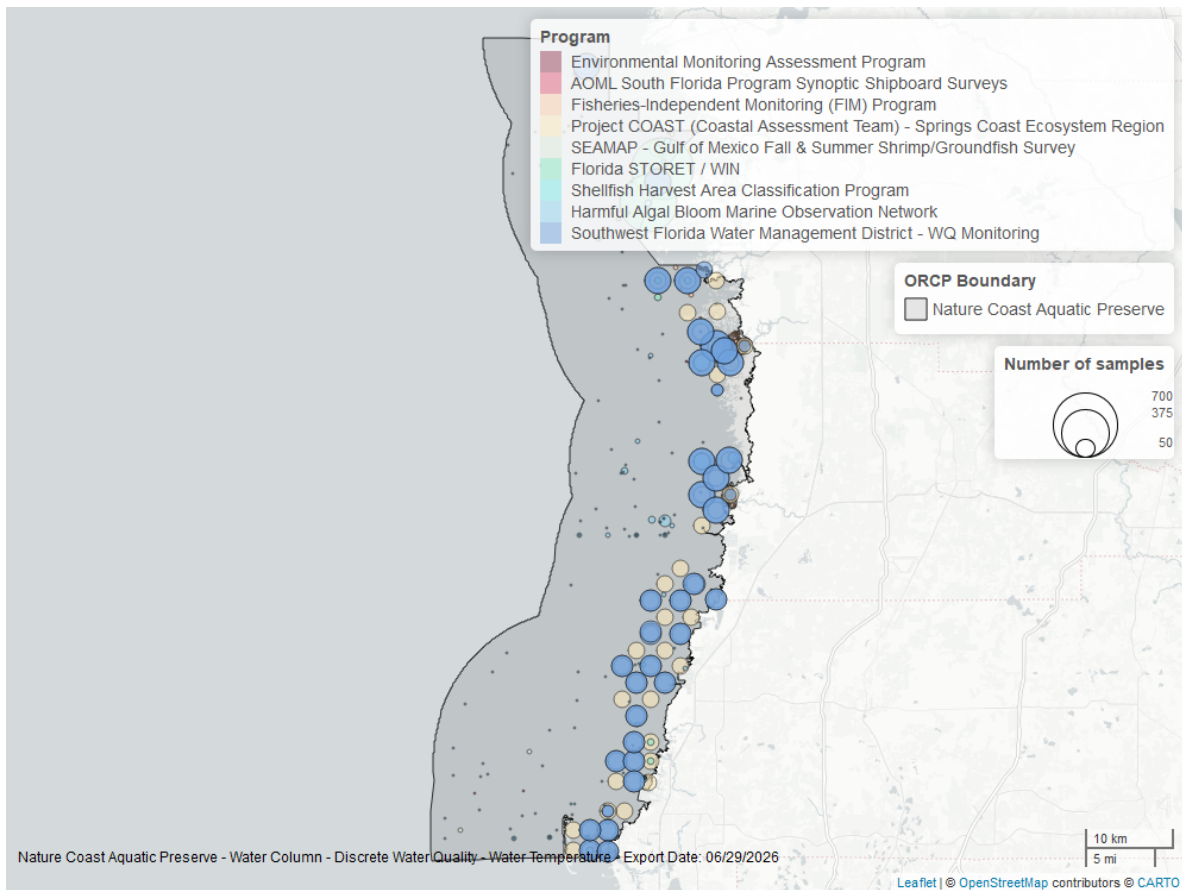


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

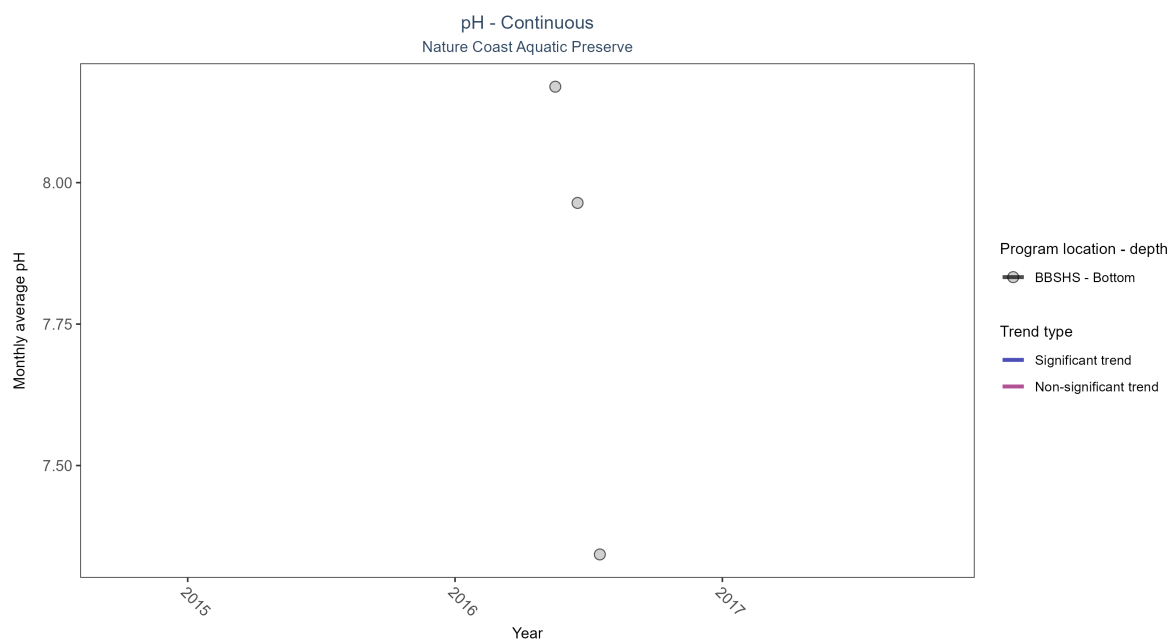


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSHS	Insufficient data	6555	1	2016 - 2016	8	-	-	-	-

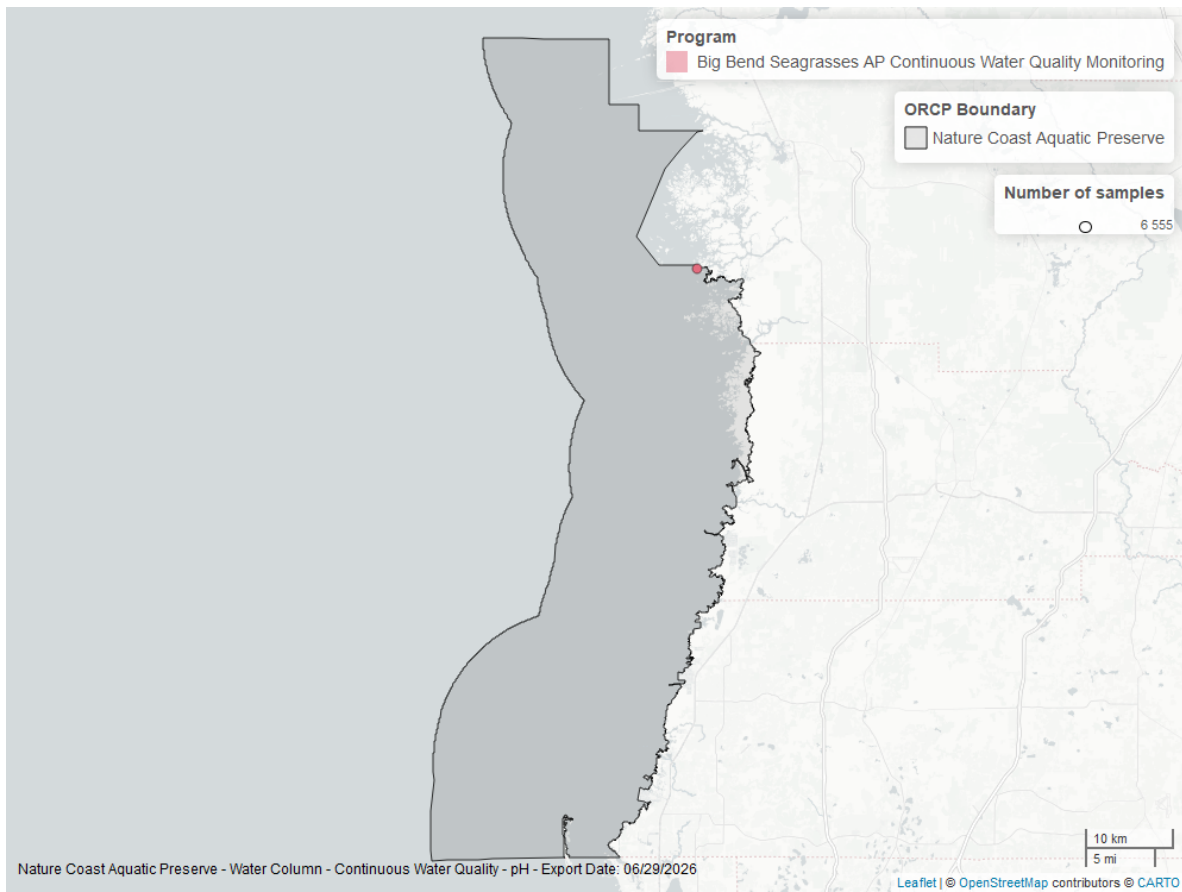


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

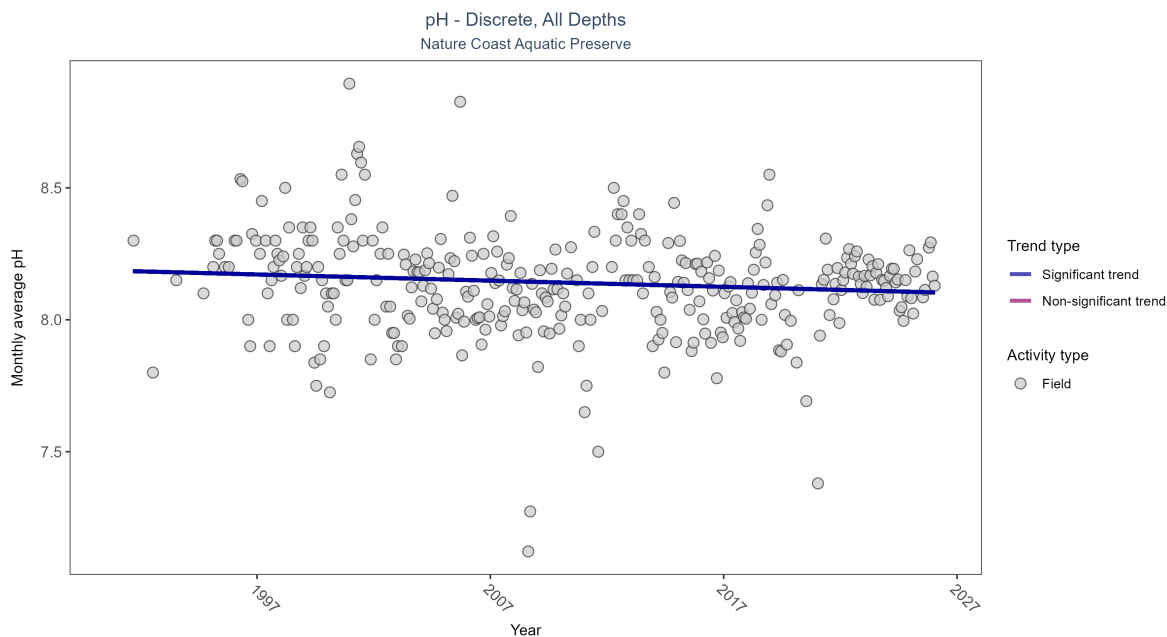


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: Monthly average pH decreased by less than 0.01 pH units per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	7985	36	1991 - 2026	8.15	-0.08656	8.18571	-0.00234	0.0232

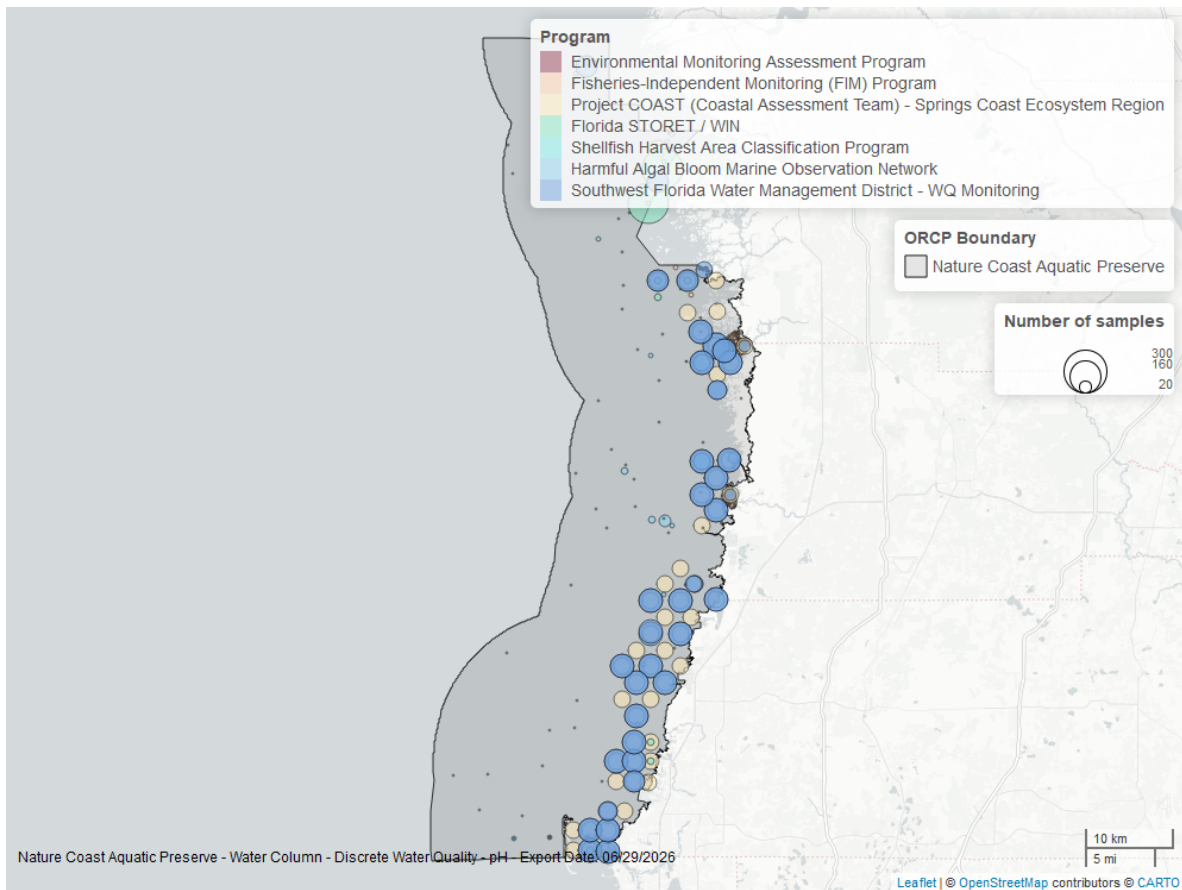


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

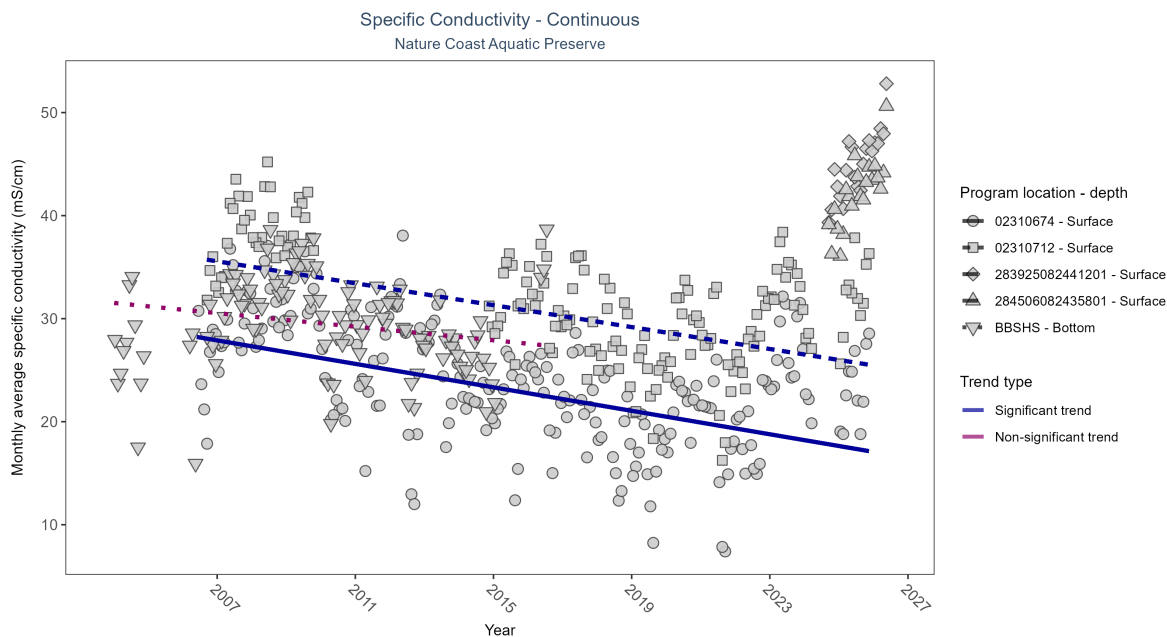


Figure 25: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 13: At two program locations, monthly average specific conductivity decreased by 0.53 mS/cm per year at one site and by 0.57 mS/cm per year at the other. No detectable change in monthly average specific conductivity was observed at one location. There was insufficient data to fit a model for two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02310674	Decreasing trend	12487	20	2006 - 2025	23.80	-0.37	28.46	-0.57	0
02310712	Decreasing trend	10064	16	2006 - 2025	32.60	-0.38	36.1	-0.53	0
283925082441201	Insufficient data	1643	3	2024 - 2026	45.30	-	-	-	-
284506082435801	Insufficient data	1642	3	2024 - 2026	42.60	-	-	-	-
BBSHS	No detectable trend	244109	12	2004 - 2016	29.73	-0.13	31.53	-0.33	0.07

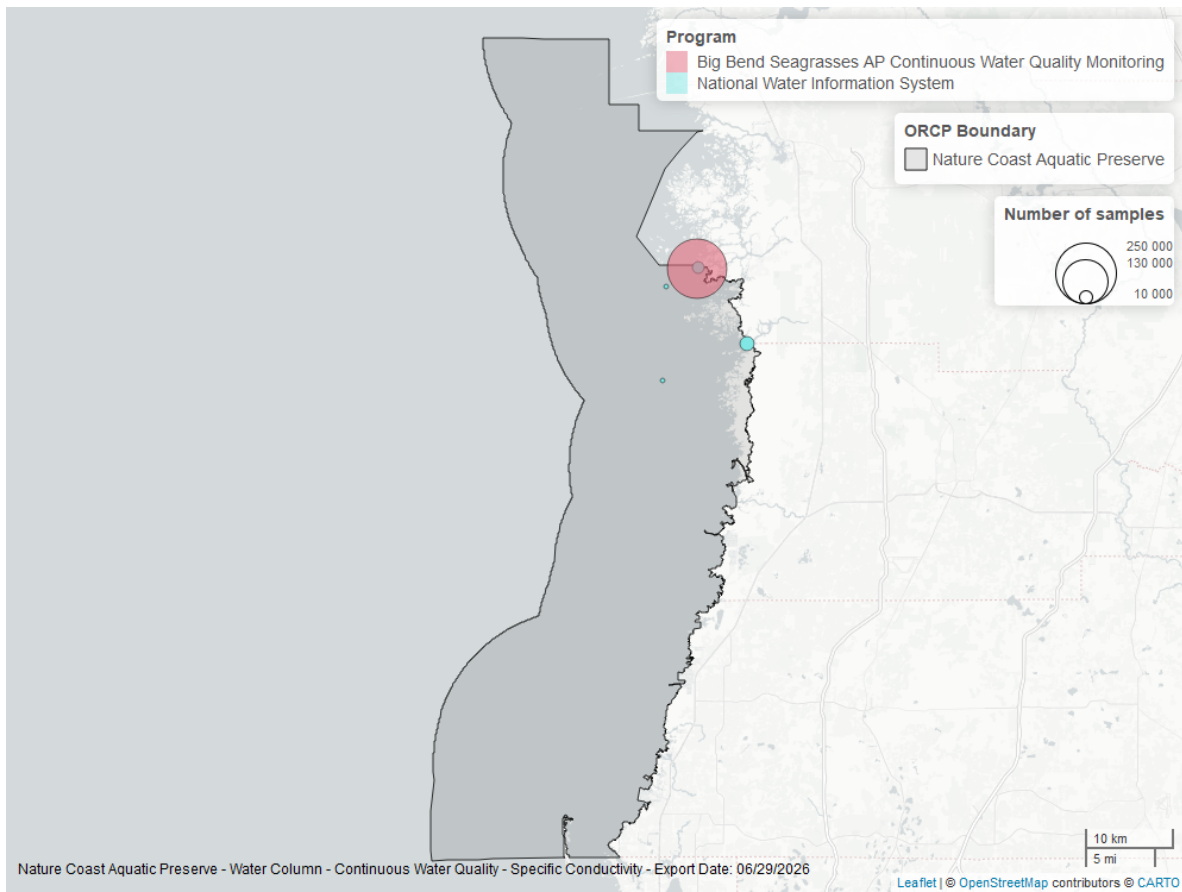


Figure 26: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

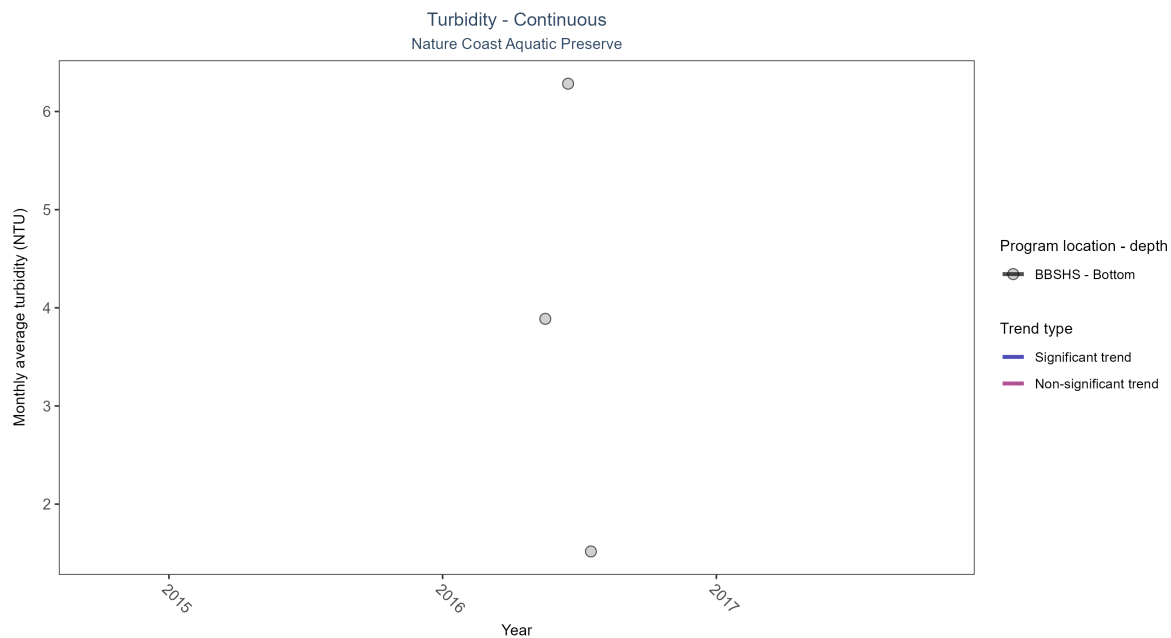


Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 14: There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSHS	Insufficient data	6368	1	2016 - 2016	2	-	-	-	-

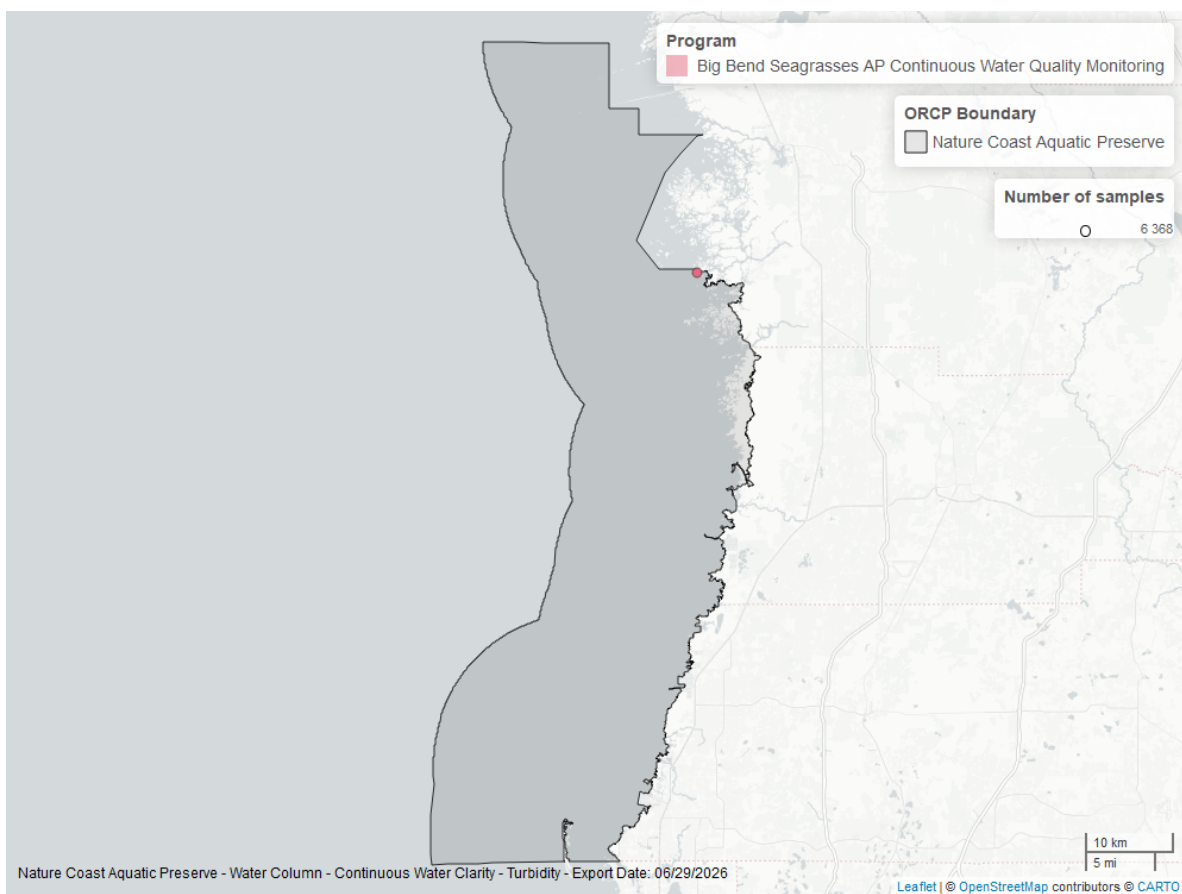


Figure 28: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

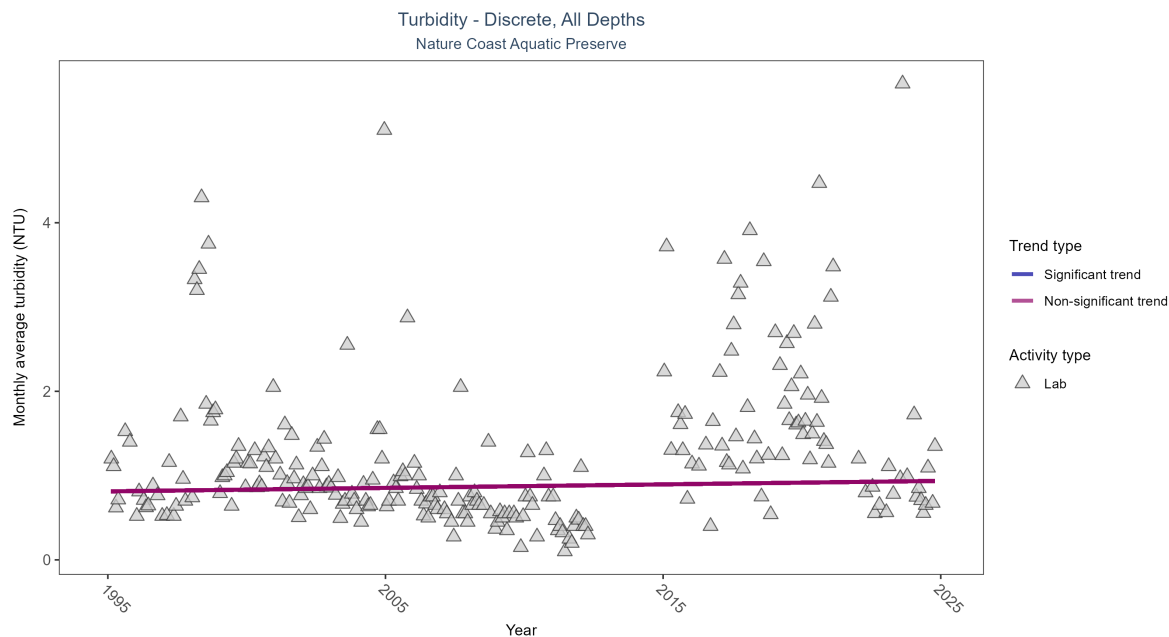


Figure 29: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 15: Turbidity showed no detectable trend between 1995 and 2024.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	1823	28	1995 - 2024	0.87	0.04785	0.81251	0.00415	0.2291

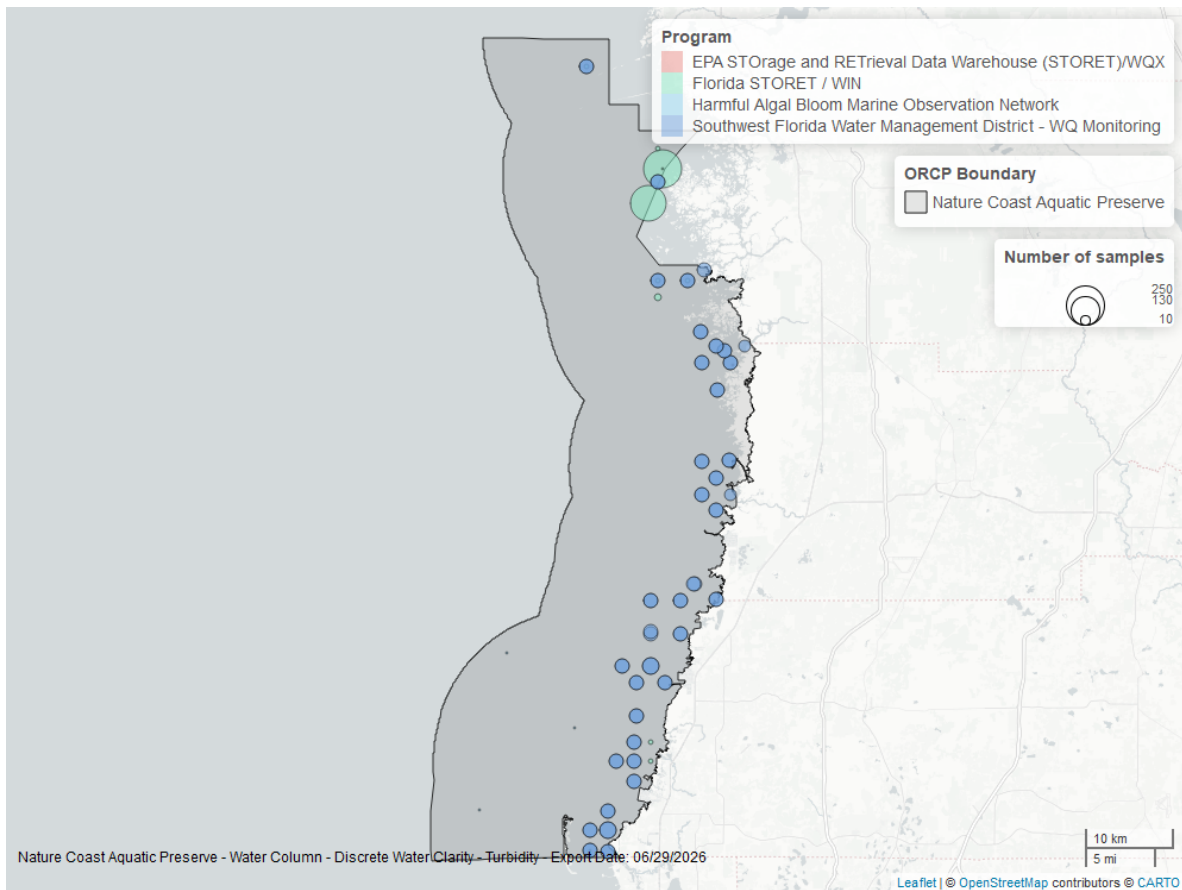


Figure 30: Map showing location of discrete water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

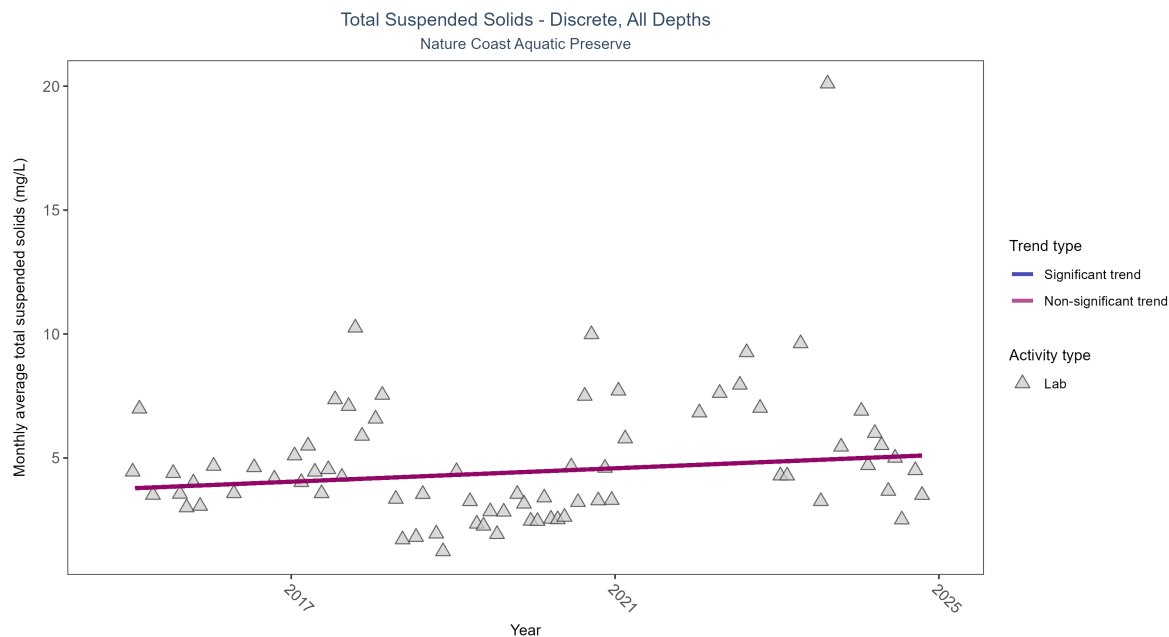


Figure 31: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 16: Total suspended solids showed no detectable trend between 2015 and 2024.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	1302	10	2015 - 2024	3.425	0.20815	3.771	0.13539	0.1146

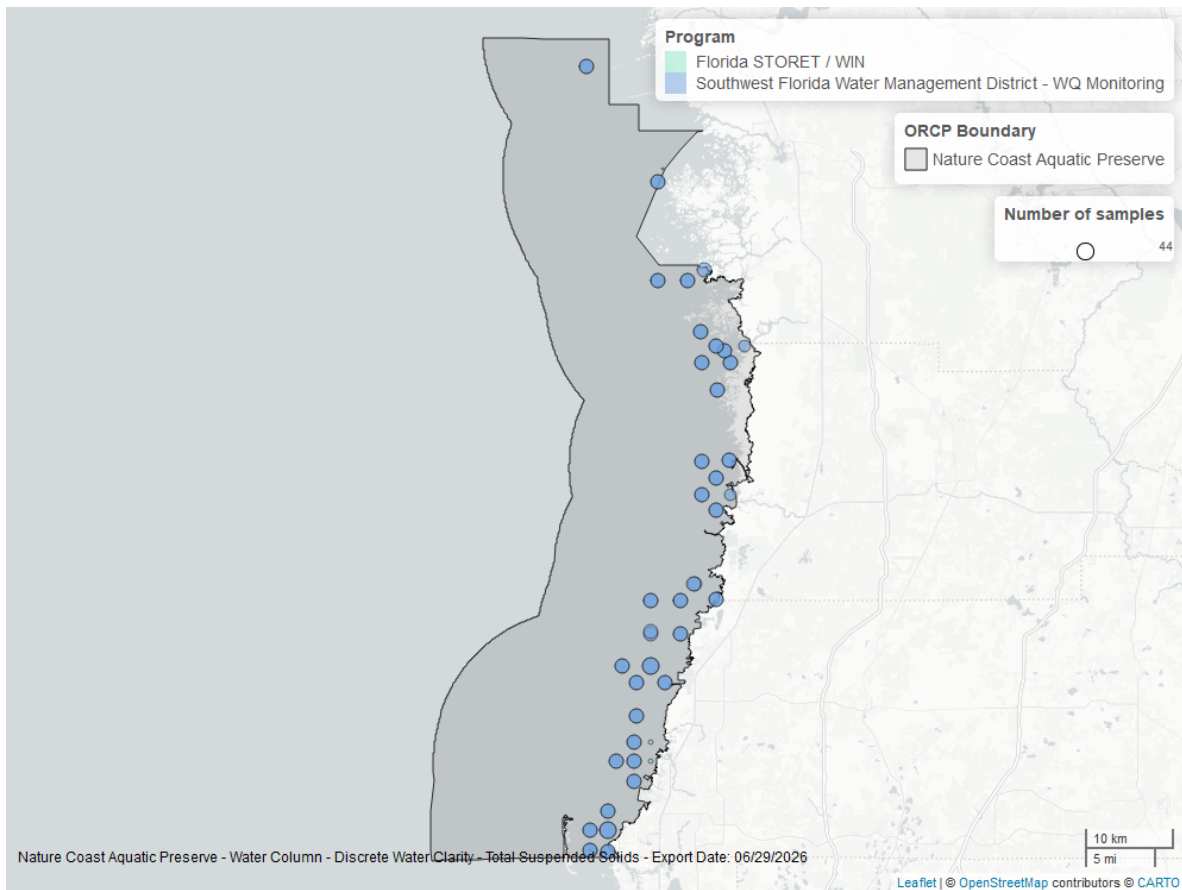


Figure 32: Map showing location of discrete water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

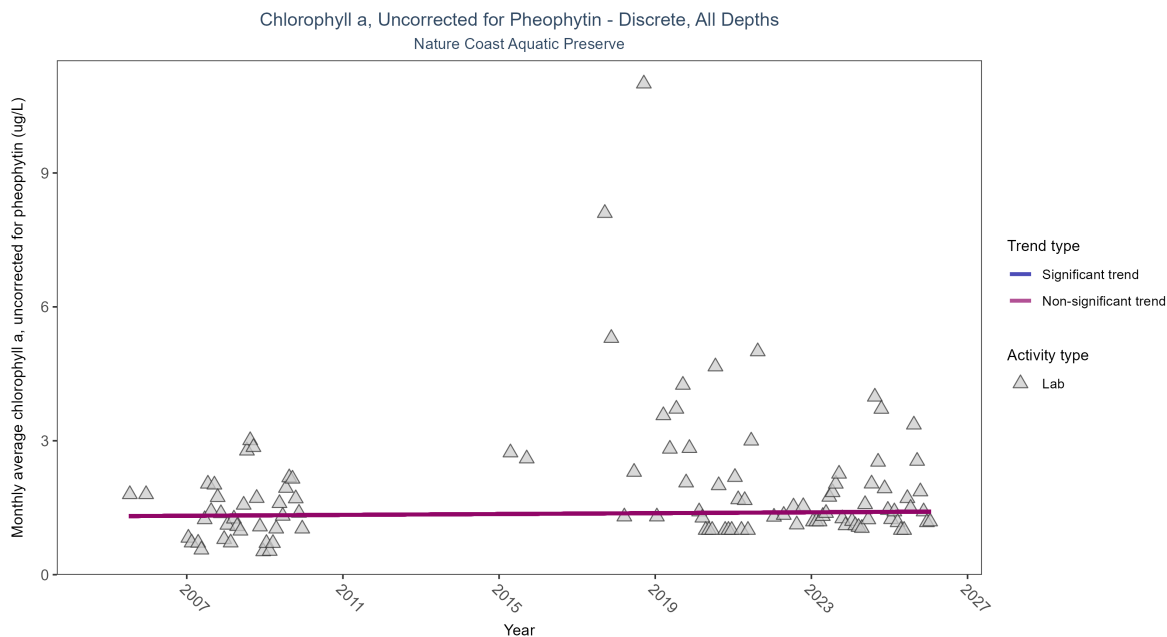


Figure 33: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Chlorophyll a, uncorrected for pheophytin, showed no detectable trend between 2005 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	3499	15	2005 - 2026	1	0.08438	1.3102	0.00485	0.3384

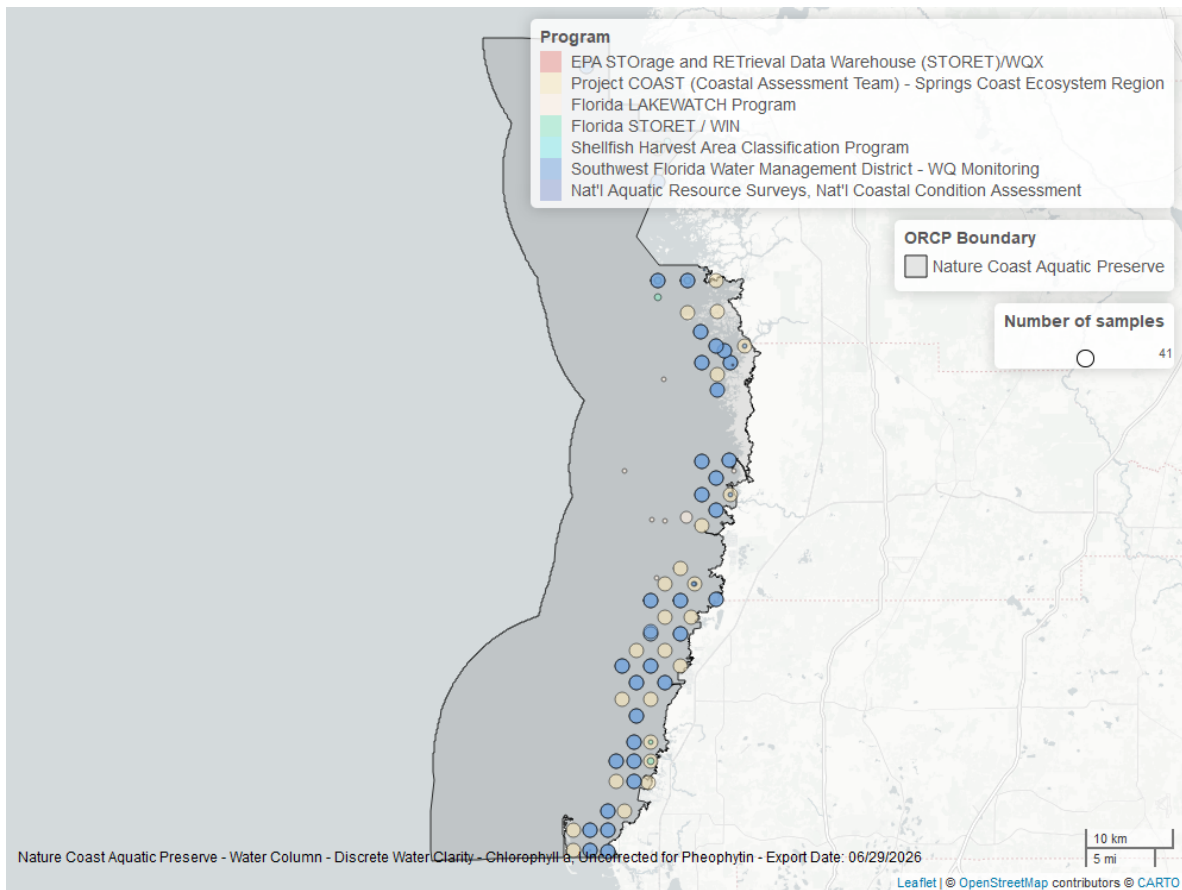


Figure 34: Map showing location of discrete water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

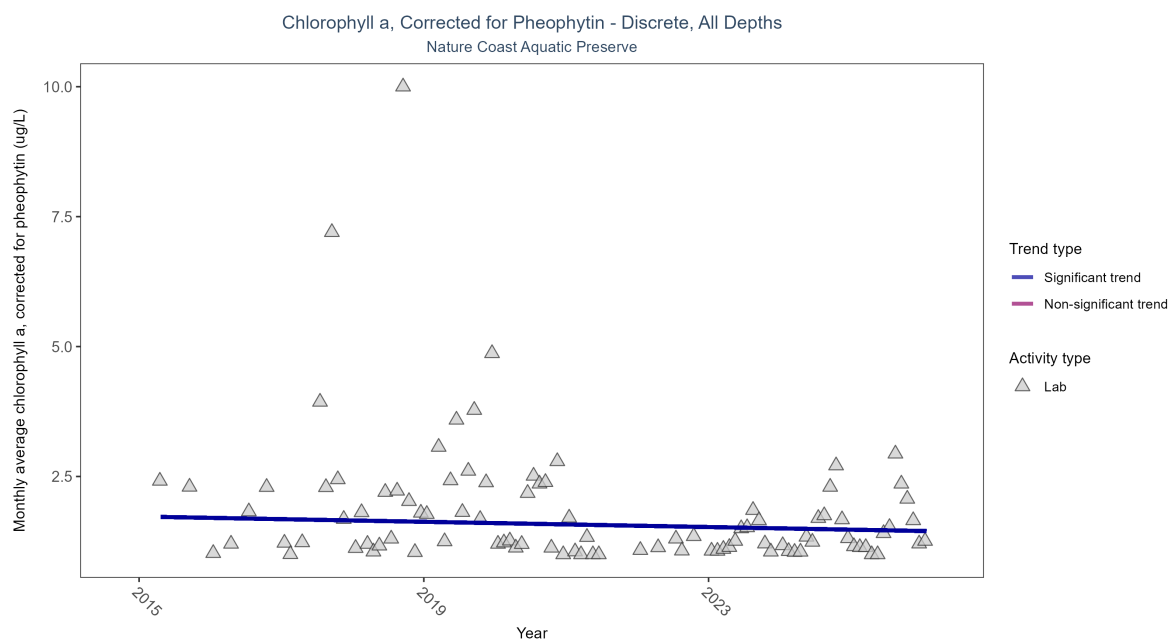


Figure 35: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 18: Monthly average chlorophyll a, corrected for pheophytin, decreased by 0.03 ug/L per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	3226	12	2015 - 2026	1	-0.20828	1.72931	-0.02527	0.0168

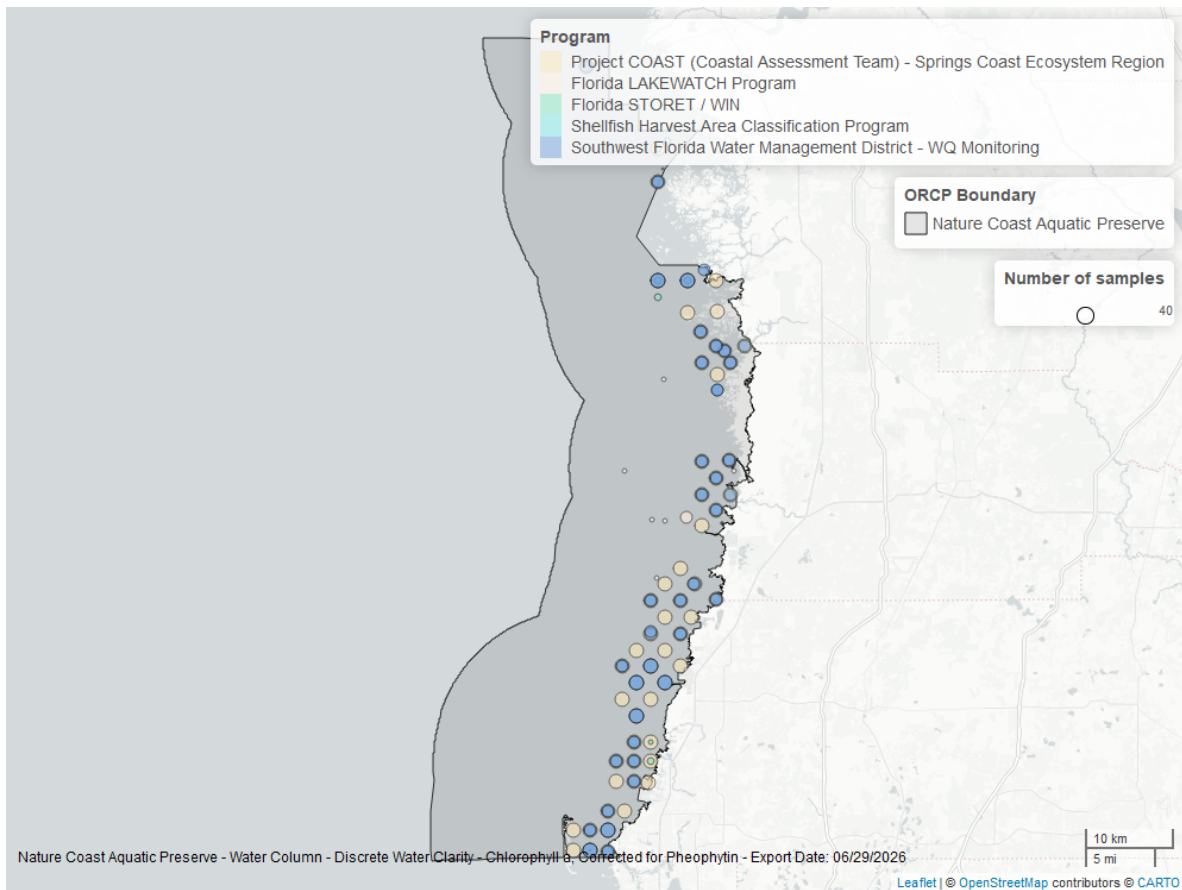


Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

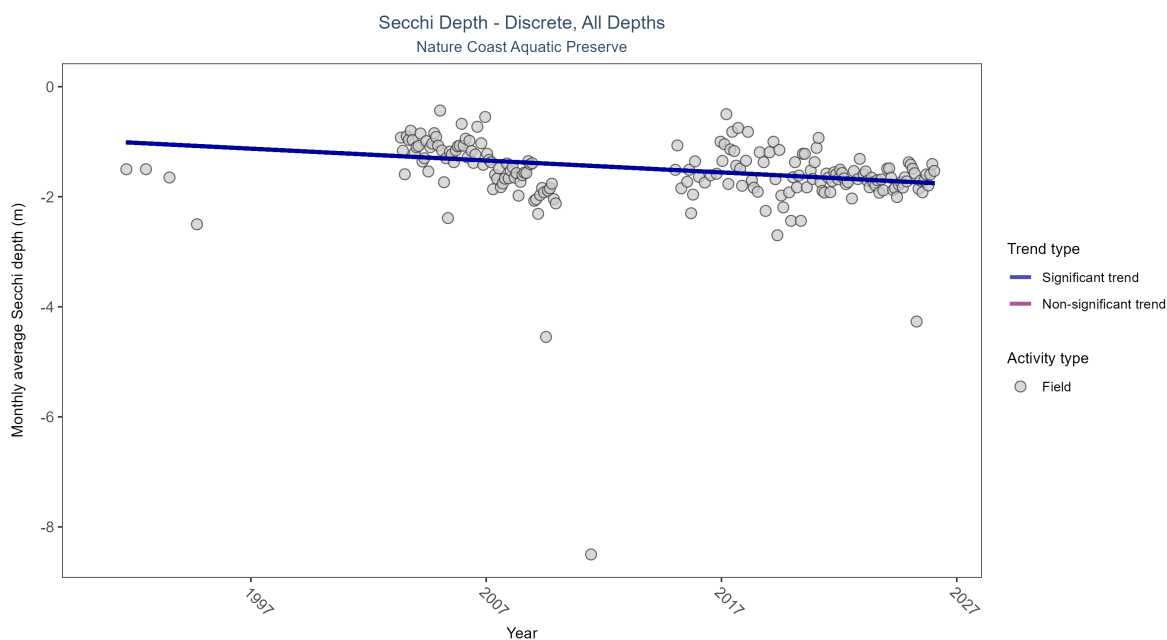


Figure 37: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 19: Monthly average Secchi depth became deeper by 0.02 m per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	5859	24	1991 - 2026	-1.5	-0.3135	-0.99653	-0.02158	0

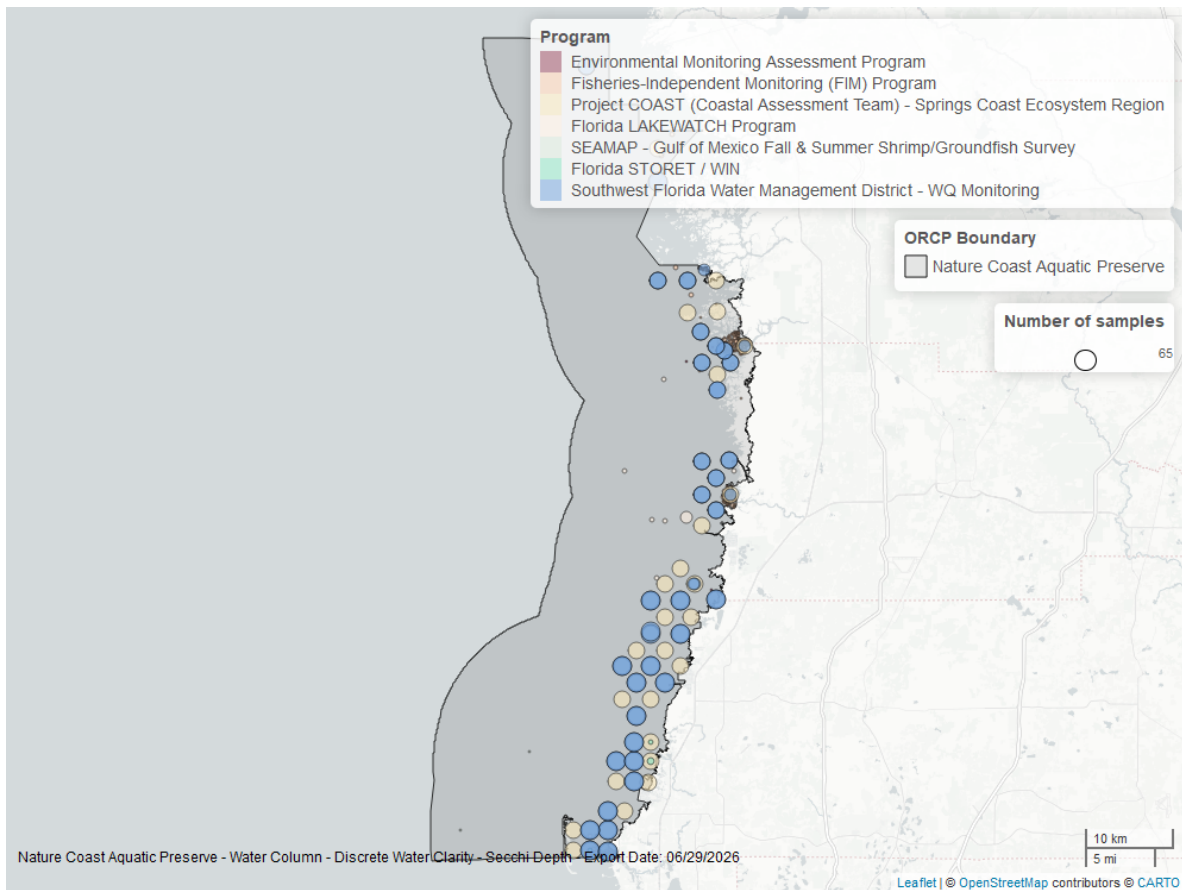


Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

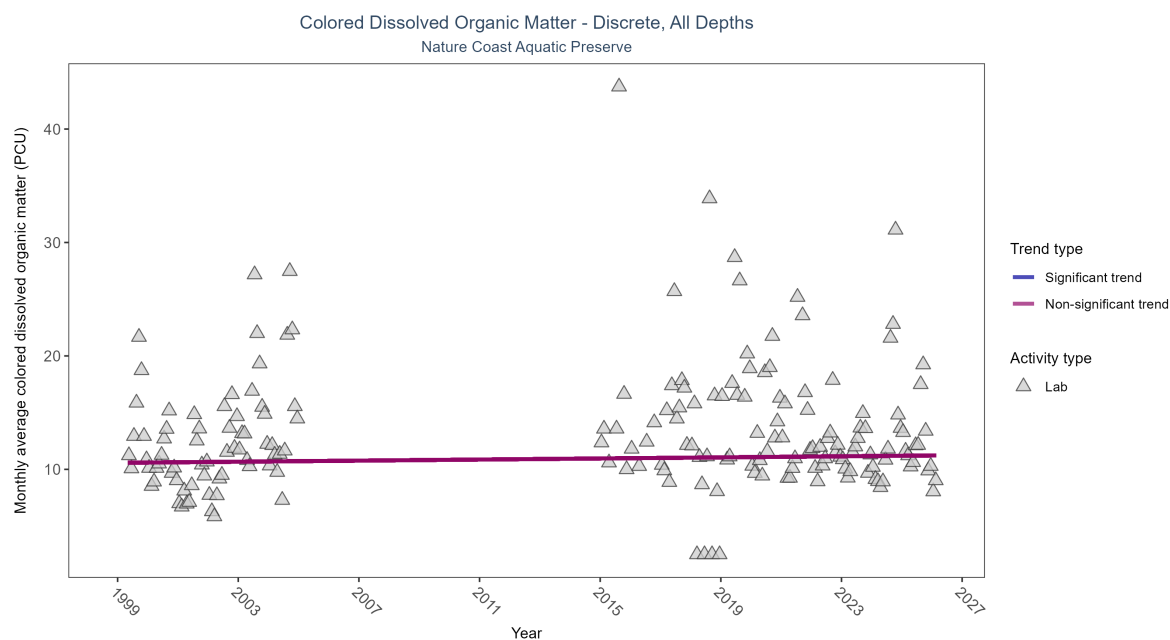


Figure 39: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 20: Colored dissolved organic matter showed no detectable trend between 1999 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	6793	18	1999 - 2026	10.389	0.07566	10.57763	0.02388	0.1863

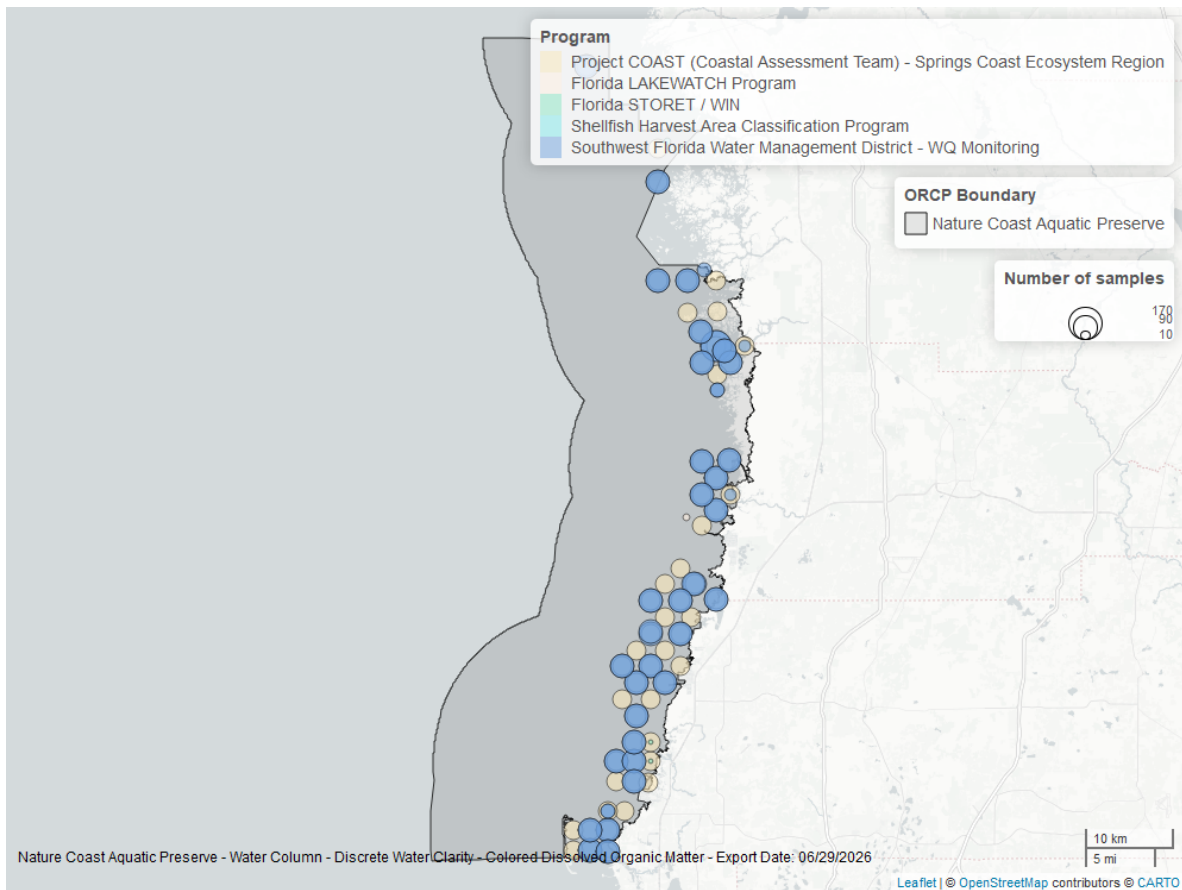


Figure 40: Map showing location of discrete water quality sampling locations within the boundaries of *Nature Coast Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.